Brokers Who Do Not Bridge: When Assessment, Not Access, Shapes Broker Advantage

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Abstract

Structural theories describe economic brokerage as an opportune but fragile position bridging gaps between otherwise disconnected actors. In that view, brokering can undermine trust or eliminate the gaps that generate its value, and brokers need supporting institutions or deeper interpersonal ties to stabilize their role. Yet many corporate actors in positions of brokerage, from Visa to Amazon, have risen in market power, profit, and prominence even as networks transformed around them. When can brokering strengthen rather than undermine a broker's advantage?

I reconceptualize brokerage as outsourced relational work, in which a broker who assumes no inventory risk constructs viable matches between parties who cannot easily find or evaluate one another. Repeated matchmaking generates an informational byproduct that the broker can accumulate: knowledge of what makes pairings successful across the market, unavailable to clients who observe only their own dealings. An agent-based model of a matching market identifies conditions under which this informational advantage emerges and persists. Brokerage can provide assessment rather than access: clients outsource matching even when they can reach counterparties themselves, because the broker's predictions are valuable. Counterintuitively, the broker can become highly central in regimes where it engages in little bridging behavior: broker centrality is a product, not a primary source, of matchmaking quality.

While the model explores the dynamics of the broker's informational advantage, I theorize how this advantage can then enable the broker to become a principal selling the resource it once intermediated or monetizing data and analytics. This is *transient brokerage*, a process that highlights a broker's growing power rather than its fragility.

I Introduction

Intermediation in economic relations—the intervention of a third party to mediate or facilitate exchange between others—can take many forms. Brokerage is one of them. In markets, brokers are intermediaries that do not assume inventory risk: they act as matchmakers, connecting others.

Brokerage is pervasive in a wide variety of markets and remarkably persistent across time and space. Brokers find themselves in an advantageous but delicate position. In structural accounts, brokers need to stand between disconnected others to have the opportunity to broker, but through their work, they densify the networks around them, potentially eroding their own advantage. Unlike dealers and wholesalers, brokers do not maintain an inventory that they need to manage and can liquidate in case of difficulty. Unlike market makers, they keep offers and price information private (Rust and Hall 2003). They operate at the intersection of social groups (Burt 1992, 2004) and rely on secret information. As a result, they often lose the trust and legitimacy needed to continue their activities. Prominent theories of brokerage have described it as an inherently fragile role that often gets "stabilized" to avoid collapsing, through the social isolation of brokers, their organizational grafting, or their capture by one side (Stovel, Golub, and Milgrom 2011).

If that is the case, then how have so many corporate actors in positions of brokerage, from Visa to Amazon, managed to rise in market power, profit, and prominence even as networks transformed around them?

Recent work by Ronald S. Burt, building on the tradition of structural-hole brokerage theory, has shown how many bridging ties are not weak or fragile, but rather strong high-trust ties forged through interpersonal history, which provide a robust and enduring advantage to the broker who relies on them (Burt 2026; Burt and Opper 2026). However, this is not the full story of broker strength.

Some brokers leverage their role to become principals rather than remain only intermediaries. They stabilize their advantage while their brokering work becomes secondary or even stops entirely. I call this *transient brokerage*, a phenomenon that current theories of brokerage do not satisfactorily predict or explain. Brokering sometimes declines or disappears because of the broker's power, not its fragility. This article explains under what conditions this happens.

This article redefines brokerage as *outsourced relational work*: the broker constructs viable matches between parties who cannot easily find or evaluate each other. This relational work generates an informational byproduct that the broker can leverage. While structural position can provide access to feed learning, each successful match between previously unconnected parties can erode the broker's structural advantage by densifying the network (Burt 2000, p. 356), but it also helps the broker accumulate knowledge of what makes pairings successful. In that way, the broker converts structural capital into informational capital through the act of brokering. This is particularly true for corporate brokers, who operate at scale and benefit from the legal privileges of the corporate form.

This article develops a theory of brokerage as outsourced relational work and explains how the resulting informational advantage can enable a transition from intermediary to principal. The article addresses three questions about brokerage:

(i) What value does a broker provide?

(ii) How does brokering alter a broker's informational and structural position?

(iii) Under what conditions can the broker convert its advantage to become a principal?

I develop an agent-based model (ABM) of brokerage in a general matching market to formalize the basis for the theory and illustrate how brokers shape and sustain structural and informational advantages through repeated matchmaking. The simulations show how several simple elements interact to produce non-obvious dynamics, complicating the expectation that brokers densify networks around them and erode their own position, and producing propositions for future empirical research. The ABM results suggest that brokerage can provide assessment rather than access, and that clients may continue outsourcing even when they can reach counterparties themselves. An experiment separating the two services shows that assessment becomes more valuable to principals as the value they derive from a match depends more on complementarity, or how well two parties fit together. When market participants leave and are replaced by new entrants at a high rate, the broker's provision of broader counterparty access becomes more valuable.

The broker ranks potential matches better than its clients. This assessment advantage comes from the information generated through its relational work. By pooling observations across clients, the broker learns how both parties' characteristics combine. By providing valuable assessments, the broker can sustain demand for its relational work, and that sustained use can increase its structural prominence. Bridging position is distinct from bridging behavior: a broker can be highly central while doing little or no bridging. The ABM results recast betweenness centrality not only as a source of brokerage opportunity, but also as a product of the kind and quality of a broker's relational work.

I build on these results to theorize how brokers can turn data obtained through their activity into a business more profitable than intermediation itself. Depending on the market in which the broker

operates, the nature of the matching problem, and the relations between the actors involved, the broker can leverage its informational advantage to convert relational work done on behalf of others into personal economic advantage. When successful, **the broker becomes a principal selling the resource it once intermediated or monetizing data and analytics**. This explains the rise in prominence of some firms acting as brokers.

The rest of this article proceeds as follows. Section II reviews structural and processual theories of brokerage and their blind spots. Section III reconceptualizes brokerage as outsourced relational work and traces how this work is transformed when performed at scale by corporate actors. Section IV presents the ABM and its results illustrating how a broker produces and sustains its advantage through this work. Section V builds on this foundation to theorize one possible trajectory that can follow from the broker advantage, transient brokerage, under which a broker can become a principal. Section VI discusses scope conditions and implications for the political economy of corporate intermediation, and Section VII concludes.

II Theories of Brokerage and Their Blind Spots

In markets where information is scarce or complex, participants are few, transactions are intricate, or institutions are poorly developed (Stovel, Golub, and Milgrom 2011), participants tend to seek the help of third parties to facilitate their transactions. Economic brokerage thus tends to emerge when counterparties are hard to find, resources hard to value, or transactions hard to structure, making intermediation work valuable. Real estate, mortgage, stock, and commodity markets are common examples of markets with many active brokers; so are markets for rare items and collectibles, over-the-counter financial markets, and many others.

Brokerage may also be helpful in circumstances where tight-knit, closed groups monopolize resources or information, and outsiders cannot easily access them (Stovel and Shaw 2012, p. 140). Brokerage can maintain or reinforce group boundaries, or it can blur or collapse them (Stovel and Shaw 2012), with consequences for social segregation, social integration, and relations between groups or communities (Galster and Godfrey 2005).

Past research has proposed two conflicting sets of definitions of brokerage (for a review, see Erikson 2013): formalist or structuralist definitions (Burt 1992; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982), and broader processual definitions (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012).

A Formalist and Structuralist Definitions

Early brokerage research examined the structure of networks and the positions of actors to identify brokerage forms, categorize them, and determine their consequences (Burt 1992, 2004; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982; Simmel 1964, 1971). Brokerage opportunities arise when there are gaps in a social network, such that some nodes are close (in degrees of separation) yet disconnected (Burt 1992, 2004; Marsden 1982). Brokerage is synonymous with open triads or structural holes: a structure in which a third actor bridges two otherwise unconnected nodes. The third actor mediates flows of resources and information and facilitates transactions between others who lack access to one another (Fernandez and Gould 1994, p. 1457; Marsden 1982, p. 202). Researchers in this tradition generally set aside interactional content to achieve greater generality (Erikson 2013). For instance, Burt (1992, p. 36) was explicit in treating "motivation and opportunity as one and the same."

Burt (1992) operationalized the structural-hole concept, proposing the measures of network constraint and effective size to quantify how well positioned an actor is for brokerage, which allowed the theory to

be tested across a wide range of settings. For instance, in a follow-up study, Burt (2004) investigated the network position of 673 managers in a large electronics company. He found that those who spanned structural holes generated ideas that senior management judged more valuable and were rewarded with better performance evaluations, promotions, and compensation. He suggested that brokers appear more creative because they can take insights that are commonplace in one group and introduce them to other people to whom they appear novel and original (Burt 2004, p. 388).

As such, brokers derive an advantage from their ability to exploit preexisting information in the network. Because of their position, brokers get privileged access to diverse ideas from across the network. They can see opportunities that others do not, and they can strategically use their position by providing or withholding information (Burt 1992) or using it to sustain their own creative endeavors. Exposed to varied perspectives, they are well-positioned to build a richer understanding, synthesize across domains or outlooks, and develop a unique vision (Burt 2004).

Defining brokerage based on structural holes, however, excludes situations that clearly appear as brokerage to a lay observer. For instance, Alice wants to sell her house and asks a real estate broker to list it. Soon after the real estate broker puts up the 'for sale' sign, Alice's neighbor Bob, whom Alice knows personally, contacts the broker to purchase the house. A broker intermediates the transaction, yet no structural hole existed between buyer and seller.

B Processual Definitions

More recently, some researchers have recast brokering as a set of relational processes (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012). Stovel and Shaw suggested that brokerage was "the process of connecting actors in systems of social, economic, or political relations in order to facilitate access to valued resources" (Stovel and Shaw 2012, p. 141; see also Stovel, Golub, and Milgrom 2011). Brokers exhibit two crucial features: they are positioned as bridges across gaps, and they help transfer resources or knowledge across those gaps (Stovel and Shaw 2012, p. 141). This definition remains constrained by social structure but requires brokerage behavior. Hence, there can be structural holes without brokerage—some bridging actors do not transmit anything across holes—but no brokerage without structural holes. In this way, it runs into the same conceptual problem as purely structural definitions.

Obstfeld, Borgatti, and Davis (2014), building on Marsden (1982), proposed a purely processual definition: brokerage is a process by which an actor "alters interaction between two or more parties in a wide variety of triadic structures" (Obstfeld, Borgatti, and Davis 2014, p. 136), meaning that he "influences, manages, or facilitates" the interaction (Obstfeld, Borgatti, and Davis 2014, p. 141). This eliminates any structural requirement beyond the need for three or more actors. It is no longer necessary for the alters to lack access to or trust in each other. There can be structural holes without brokerage, and there can also be brokerage without structural holes: brokerage behavior can, and often does, occur in dense networks.

Previous work on brokerage, particularly in the processual tradition, has also identified categories of brokers depending on the nature of their activities. Middleman, transfer, or conduit brokers facilitate economic transactions without necessarily altering the structure of a local network, like stockbrokers (Stovel and Shaw 2012, p. 146; Spiro, Acton, and Butts 2013; Obstfeld, Borgatti, and Davis 2014, p. 141). Catalyst, moderator, or matchmaking brokers focus on creating lasting ties, like headhunters and party hosts (Stovel and Shaw 2012, p. 146; Obstfeld, Borgatti, and Davis 2014; Spiro, Acton, and Butts 2013). Finally, *coordinators*, such as middle managers or disaster response teams, work to transform existing ties by reconfiguring local networks of interdependence (Spiro, Acton, and Butts 2013, pp. 131–132).

While definitions based on structural holes were very restrictive, Obstfeld and co-authors' processual definition is sweeping. For instance, they suggested that, under their definition, a manufacturing

firm acts as a broker when it buys raw materials from an alter, transforms them into a product, and sells the finished product to another alter (Obstfeld, Borgatti, and Davis 2014, p. 145). Most triadic or polyadic relations, in this account, imply brokerage. Indeed, many descriptions of brokers in the literature are so broad as to invite the charge that a concept that explains everything explains nothing. The most varied social facts have been described as acts of brokerage: mediating conflict (Obstfeld, Borgatti, and Davis 2014), reselling stolen property (Steffensmeier 1986), advocating on behalf of someone, making sense of the world for someone, and helping connect epistemic communities (Stovel and Shaw 2012, p. 140), to name only a few.

C Shortcomings of Existing Frameworks

Work that relies chiefly on the structural holes idea generally makes clear that transferring knowledge and ideas is a key element of brokering and of the broker advantage. Yet the information under consideration is generally preexisting and ambient, not generated by brokering itself (Burt 2004). Moreover, the structural account leaves the actual work of brokering largely implicit. Structural position indicates places where brokerage is possible, but not whether anyone is doing the work of brokering—work for which structural holes are neither necessary nor sufficient.

Meanwhile, broad processual definitions blur the lines between vicarious and personal relational work. They include under the same umbrella cases where the intermediary deals with others on its own account, and cases where the intermediary works on behalf of others. As such, this perspective does not examine the informational byproduct that results from repeated vicarious work. Information is often simply listed among the resources that can be transferred through a broker, along with material or financial resources (see, e.g. Obstfeld, Borgatti, and Davis 2014, p. 141; Stovel and Shaw 2012, p. 141).

Neither perspective offers a cohesive theory of the role of information in brokerage, even though information shapes every stage of the brokering process. Before brokerage, the distribution of information shapes the opportunity or need for it: brokerage arises partly out of informational inequality, and the broker exploits what it knows that one or both sides do not. During brokerage, the transmission of information is part of the brokerage act itself, as the broker shares enough to make a match viable while withholding enough to preserve its advantage. After brokerage, the distribution of information has been altered, and the work has generated a byproduct the broker retains, which may shape the broker's opportunities. The circulation of information is at once a determinant of brokerage, a component of the work of brokering, and a consequence of brokerage.

This article reconceptualizes brokering as outsourced relational work and builds on this theoretical move to explain how brokerage sometimes weakens or disappears because a powerful broker captures resources or encloses data to become a principal.

III Outsourced Relational Work and the Broker Advantage

Social actors engage in *relational work* when they try to interpret and match social ties, economic transactions, and media of exchange, to create viable matches that will shape the outcome of their interaction (Zelizer 2012, p. 151, 2005; see also Bandelj 2020, p. 255). These matches and their shared meanings form distinct *relational packages*. For instance, in a marketplace, actors trading directly with one another draw on cultural knowledge to reach a shared understanding of their situation and the 'rules of the game,' and to agree on terms, including price (Block 2012, p. 137).

For the broker's clients, brokerage is *outsourced relational work*. For the broker, brokerage is *vicarious relational work*. In economic markets, this outsourced relational work is usually performed at scale by a *corporation*. This is more than a question of definitions: it shapes the broker's opportunities, its strategic advantage vis-à-vis its clients, and its trajectory.

A From Personal to *Outsourced and Vicarious* Relational Work

In brokered transactions, brokers take on the work of constructing a shared reality for limited purposes. The core of what brokers do is relational matching (see Zelizer 2012). Brokers are characterized by the relational work they do *on behalf of others*.

For the broker, this vicarious work requires another layer of relational work to negotiate brokerage relations with its clients. Brokerage corresponds with a distinct relational package. Not all intermediation is brokerage. In the relational package of economic brokerage, the broker performs a matching service for others, usually for a fee, commission, or retainer. Brokers bundle various services and amenities as part of that package: they take on the efforts of search and negotiation, they provide access to otherwise inaccessible counterparties, they share with their clients their expertise and the benefits of the private information they hold about quality, pricing, and availability, and they document and facilitate transactions, which often require logistical, bureaucratic, or regulatory work.

The middleman, catalyst, and coordination brokerage described in Section II are not exclusive categories of brokers, but rather, components of the work of brokering, which involves a bit of each in varying proportions. An important part of brokers' relational matching is to facilitate interaction (middleman brokerage), but this work also directly impacts interpersonal relations and local networks, often in an enduring way, by creating new and lasting ties (catalyst brokerage) or by reorganizing local networks (coordination brokerage).

Brokerage inherits the dual character of relational work, which can both foster and reduce informational asymmetry when constructing shared understandings while maintaining strategic opacities (Block 2012, p. 138). In performing relational work on behalf of others, the broker simultaneously clarifies and obscures: it transmits enough information to make a transaction legible and a match viable and conceals enough to justify its fee and preserve its advantage. The same opacity can do social as well as economic work. Rossman (2014) shows how brokerage functions as obfuscatory relational work that manages the taboos around disreputable exchange. Outsourcing relational work provides cover to parties in morally fraught matches. What the broker withholds is also the raw material of its later advantage: information that it withholds, accumulates, and, under the right conditions, eventually monetizes.

Vicarious work relates to entities or resources that are *not* under the broker's personal ownership and control, and the shared understanding of the parties is that the broker is *not a principal* in the brokered transaction (see, e.g. Stovel, Golub, and Milgrom 2011, p. 21326). Brokers generally have limited personal involvement in brokered interactions beyond their role as facilitators. The broker is a supporting actor, not a main character. Intermediaries who have full control over traded resources occupy a very different position in terms of risk, power, and vulnerability.

Another way to put it is that in economic markets, brokers are intermediaries that *do not assume inventory risk*. That is, if Bob helps Alice and Charlie conclude a sale, Bob is a broker. If Bob buys from Alice and eventually leases to Charlie, he is not. For instance, a real estate broker using her market knowledge to buy houses in up-and-coming neighborhoods to rent to families no longer acts as a broker. Instead, the relational work she performs is personal: matching herself to sellers and renters.

B From Individual to *Corporate* Relational Work

The definition above applies naturally to describe individual brokers working with individual clients, such as a real estate agent, an insurance broker representative, or a matchmaker. Much of the work of brokerage, however, occurs at the corporate, organizational, or institutional level, where brokers are firms, and clients may also be.

To understand what happens when relational work is outsourced to corporations, we must first discuss how corporations perform relational work in the first place, for themselves as organizational social actors. Relational work scholarship initially focused on the intimate sphere and interpersonal relationships (for a review, see Bandelj 2020). But relational work also takes place in the economic sphere and at the organizational level. The relational work tradition has acknowledged the organizational question without fully resolving it. Bandelj (2012) noted that relational work between representatives of organizations solidifies into macrolevel structures, while Zelizer (2012, pp. 156–157) catalogued instances of relational work by firms in the field of labor relations. Bandelj's (2020) review of "scaling up" relational work surveyed studies operating at the organizational level, but they mainly focused on what people do with and to each other within organizations, not on the corporate form itself as a vehicle for relational work.

Firms must conduct relational work to manage contingent, fleeting ties and longer-term relationships with workers, consumers, suppliers, lenders, investors, bureaucrats, and so on. The properties of the corporate form matter to this work and make it distinct. The firm is not simply a site where interpersonal relational work occurs; nor is it a singular, scaled-up actor that performs relational work as an individual does.

Firms are groups of people in designated roles, following norms and practices, supported by procedures, interfaces, and infrastructures, and doing the everyday work of managing the firm's relationship with its environment "at the coalface" (Gray and Silbey 2014; Huising and Silbey 2018). The mining metaphor directs attention to the frontline of the organization, where people enact its dealings with the outside world and translate organizational imperatives into routine practice. Relational work, like everything the corporation does in the world, is enacted through the people, procedures, infrastructure, and algorithms working at the coalface. Because of its collective and distributed nature, it is never visible in its entirety to any of the people involved.

Most firms are also legally organized as corporations. The corporation is a political entity with emergent capacities for coordination, information processing, and strategic action that exceed the sum of its parts (Ciepley 2013; Farrell and Newman 2023). There is a fundamental asymmetry between corporate actors and natural persons (Coleman 1982, 1990). The corporation aggregates information across thousands or millions of interactions, while each natural person sees only their own. The corporation can persist indefinitely, while individual relationships are transient. The corporation coordinates strategically across its full portfolio of relationships, while each counterparty acts in isolation. As a result, when a corporate actor interacts with natural persons, the corporate actor has structural and informational advantages.

These asymmetries are not accidental byproducts of size: they are legally constituted features of the corporate form. The corporation is one of the core modules of the legal code of capital (Pistor 2019). Corporate law endows the firm with durability that outlasts any individual relationship, priority that ranks its claims above those of other parties, and universality that makes those claims enforceable against the world, so that it can become a legal technology for concentrating and protecting accumulated advantages and leveraging them for productive purposes. These are the conditions under which corporate relational work takes place: it is inherently asymmetric, because of the different scales and material and informational resources of corporate and natural persons.

C Outsourcing Relational Work to Corporations: The Corporate Brokers

When a client outsources relational work to a corporate broker, the corporate broker performs that work through people and practices at the coalface, each working in some small way to define relational packages, perform appropriate matches, and shape information flows. The corporate form reorganizes vicarious relational work, distributes it across roles, and embeds it in routines and procedures. To

orchestrate this, corporate brokers develop an organizational apparatus comprising internal and client-facing roles, information architectures and databases, contractual templates, and interfaces.

Saying a brokerage firm does relational work on behalf of its clients means that in sales meetings, in contract negotiations, in the design of user interfaces, in the defaults embedded in terms of service, and so on, people relying on organizational procedures are working to define the features of its clients' relationships, what transactions are appropriate to them, and what media may govern the exchanges. As with its own relational work, the corporation has a capacity for coordination, information aggregation, and temporal persistence that makes the vicarious relational work of corporate brokerage qualitatively different from what any individual could accomplish alone.

The corporate broker's relation with its clients is structurally asymmetric. With human clients, corporate and natural persons bring fundamentally different legal, informational, and temporal architectures to the relationship. Even with corporate clients, however, the asymmetry remains: the broker coordinates vicarious relational work strategically and benefits from cross-market information, while the client encounters it piecemeal. The matching of relations, transactions, and media moves from a pure negotiation between principals with comparable standing, who learn from each transaction, toward an administered process in which the corporate broker sets the terms, explicitly or behind the scenes, and captures the information that each pairing generates.

The data infrastructure of the corporate broker includes data taxonomies, client categories, risk models, algorithmic sorting procedures, and so on. This infrastructure does not merely facilitate transactions but actively constitutes relational packages by determining what transactions are possible, and what categories they should fall into. Classification systems and information infrastructures, like those of the corporate broker, are not neutral descriptions of the world; they actively organize social relations (Bowker and Star 1999; Fourcade and Healy 2024). The corporate broker's information infrastructure, much like a corporate rating agency's credit-scoring system or a platform's matching algorithm, sorts counterparties into categories the broker defines and uses those categories to allocate opportunities according to the broker's objectives. The impact of the broker's information infrastructure is magnified because the corporate broker works at scale, while individuals cannot usually design categories that simultaneously structure millions of relationships.

D The Broker Advantage

Why do people and firms outsource their relational work to brokers? To have the opportunity to broker, a participant in a matching market must hold either a *structural* or an *informational* advantage. A structural advantage arises from network position and can be used to provide *access*: the broker bridges by bringing together parties that were not previously connected. An informational advantage supports *assessment*: the broker evaluates prospective matches better than its clients can. The broker's service is therefore valuable when it helps clients reach counterparties they could not otherwise find or better evaluate counterparties already available to them.

Assessment involves two components: (i) evaluating the individual quality of the goods or counterparties involved and (ii) evaluating their complementarity, that is, the quality of the specific pairing or relational package being considered. Individual quality and complementarity each contribute to the value that a match generates for its parties.

In some markets, for example on the New York Stock Exchange, complementarity does not matter much or at all. In markets with limited liquidity, no centralized trading venue, or nonfungible goods, however, like the labor market, the market for art, or over-the-counter financial markets, complementarity becomes very important.

Individual quality and complementarity components correspond to two channels for the broker's informational advantage: *attributional*, about evaluating how parties' attributes translate into

individual quality, and *relational*, about evaluating which counterparties best complement each other. Both are self-reinforcing, as the broker accumulates the knowledge required for assessment through the relational work it performs, and that learning makes it better at the work. The broker observes outcomes from many pairings across many principals, while each principal observes only its own. This gives the broker both a cross-market view and higher data volume.

The broker's information work can be informal and internal, as with a real estate broker who knows the local market, has an intuition for the right asking price, and holds social ties to likely buyers. It can also be formal and bureaucratized, as with a corporate mortgage broker that follows tables and formulas to determine which interest rate may produce successful matches between borrowers and lenders. The latter records and retains what the former holds in personal notes or experience.

Important models in the economics literature have focused on the attributional channel and characterized the broker's role primarily as quality certification (Li 1998) or expert screening (Bethune, Sultanum, and Trachter 2022): the broker identifies which goods or counterparties are high quality. In that view, the broker is an appraiser whose cross-market experience helps it assess the individual quality of counterparties more accurately than individual principals can. In contrast, the relational-work view of brokerage suggests that the broker's value lies also, and often primarily, in understanding complementarities between counterparties and shaping relational packages accordingly. The broker is a relational worker and a matchmaker whose advantage comes from the ability to make pairings and to know which pairings will likely succeed, an ability that improves with repeated matchmaking.

The model that follows illustrates the basis for the theory: a repeat-player broker observes outcomes across clients and uses the resulting information to organize subsequent matches. The model portrays how the broker's work, outsourcing rates, and network topology can evolve together.

IV How Brokers Accumulate Informational and Structural Advantage: A Formal Illustration

A Model and Methods

I use an agent-based model (ABM) to formalize the theory. The model is an illustration, not a test of the theory; it serves as a bridge from theory to testable propositions. Complete pseudocode and specifications appear in Appendices A and B. The public Git repository (`https://github.com/m-laprise/brokerage-abm`) contains the Julia code and scripts needed to reproduce the simulations, statistics, figures, and manuscript text.

Market and match output. Principals are the market participants seeking partners. Many principals and one broker occupy an undirected network. Each principal has a fixed vector of observable characteristics, which defines its type. At baseline, types are drawn around a smooth curve in an eight-dimensional space of observable characteristics (Supplementary Figure S1). For a given pair, the true noise-free portion of the output that will be generated by their match (hereafter, the underlying match value) is fixed by the two principals' characteristics and the matching environment drawn at the start of each simulation run. This value excludes match noise, search costs, and broker fees. Underlying match value combines two components: individual qualities and complementarity (Appendix B §2). Each principal's individual quality is a function of its own characteristics and the mapping from characteristics to quality is fixed and shared across the market. The individual-quality component is the average of the two principals' individual quality. Complementarity depends on how their characteristics interact.

When underlying match value depends only on individual quality, all principals have the same ranking of shared candidates by underlying match value. With complementarity in the equation, these rankings can differ across principals (Supplementary Figure S2). Predicting individual quality corresponds

to attributional assessment; predicting complementarity corresponds to relational assessment (see section III.D).

Before they are combined, the individual-quality and complementarity components are centered and scaled to have equal average within-market variances across the calibration markets. They receive weights $\sqrt{\rho}$ and $\sqrt{1-\rho}$, respectively. On this calibration scale, the individual-quality component accounts for a fraction ρ of the two weighted components' summed marginal variances, and complementarity accounts for $1-\rho$. At $\rho = 0$, underlying match value depends only on complementarity; at $\rho = 1$, it depends only on individual quality. The parameter δ controls the difficulty of learning complementarity by varying how sharply complementarity changes across an arbitrary boundary in type-pair space: it multiplies complementarity by $1+\delta$ on one side and $1-\delta$ on the other. Larger values make otherwise similar pairs on opposite sides of the boundary differ more in underlying match value. Changing ρ and δ changes the shape of the distribution of match values.

The normalization sets the average within-market variance of underlying match value to one in independently generated calibration markets, providing a common reference scale across simulations. The normalization coefficients remain fixed over time as turnover occurs and principals enter and exit, keeping the matching function stable. The range of underlying match values across pairs varies with ρ and δ. The range is wider in complementarity-dominated regimes. Increasing the difficulty widens it further by amplifying interactions for some of the pairs while suppressing them for others. This concentrates variation among the amplified pairs and leads to more extreme values even after normalization.

A different total net output to principals across such regimes cannot be cleanly attributed to a difference in matching performance. Consequently, when comparing regimes with different matching compositions or difficulties, I report differences in the share of principals' net output from brokered or self-search matches. I compare net output levels between service versions within each regime. Because δ affects only complementarity, it has no effect when $\rho = 1$. Describing variation in underlying match value across pairs requires more dimensions when complementarity contributes than when underlying match value depends only on individual quality, particularly when difficulty is high (Supplementary Figure S3).

When a pair matches, its realized match output equals its underlying match value plus an independent, mean-zero random shock. This shock represents match-specific variation that neither the principals nor the broker can anticipate.

Search, matching, and learning. Each period, principals seek pairwise matches either through self-search or through the broker (Appendix A §§5–7; Appendix B §§6–7). They choose one channel using satisfaction with the returns on their past matching requests, after search costs or broker fees. Self-search considers existing neighbors and a small random sample of other principals. The broker considers principals in its standing roster (a persistent list of contacts), and those currently using its services.

Principals and the broker accumulate observations and learn in parallel. After every completed match, each principal records its partner's characteristics and the realized output, whether the match was arranged through self-search or through the broker. The broker also records both parties' characteristics and realized output from every match it mediates, pooling observations across clients. It does not observe matches arranged entirely through self-search.

Each principal uses its own observations to estimate the underlying value of pairing with a candidate, using that candidate's characteristics. The broker uses its pooled observations to estimate this value from both parties' characteristics. These estimates are the principals' and broker's predicted match

values. In the main variant, each principal has its own neural network, and so does the broker. In another variant, these predictions use ridge regression.

Predicted match value determines how candidates are ranked. For previously matched neighbors, self-search uses the pair's average past realized output as its prediction. Candidates with equal scores are ordered uniformly at random. Principals make offers to the highest-ranked candidates with whom the predicted value of a match exceeds the reservation threshold. All offers enter a common market. A completed match creates a direct tie that persists until either principal exits. At the end of each period, every principal exits independently with probability η (the turnover rate) and is replaced by a new entrant.

Measurement. The outsourcing rate is the share of requested match positions routed through the broker. A brokered placement is classified as access when it creates a direct tie between previously unconnected principals and as assessment when it reuses an existing tie. The access fraction is the share of brokered placements classified as access. Net output per principal is total realized match output, counting each completed relationship once, minus all self-search costs and broker fees, divided by the number of principals. Self-search costs include costs incurred on unsuccessful searches. Broker betweenness centrality measures the share of shortest paths between principals that pass through the broker (Brandes 2001; Freeman 1977). Supplementary Figures S5–S6 report the corresponding structural comparisons using Burt's constraint and effective size.

Prediction quality is evaluated each period using randomly sampled potential partners. These evaluations do not update either model. Each predicted match value is compared with a pair's underlying value. Rank correlation is Spearman's correlation between the candidate ordering produced by predicted match values and the ordering produced by underlying match values. The ranking advantage is the broker's rank correlation minus the principals' rank correlation (Appendix B §8).

Simulation design and analysis. At the baseline, $N = 1{,}000$ principals interact for $T = 500$ periods with $\rho = 0.5$, $\delta = 0.5$, and $\eta = 0.02$ (Appendix B §9). The one-at-a-time sweeps and four two-parameter grids are described in the supplementary section "Simulation design and analysis" (Tables S1–S2). I refer to each unique model configuration as a regime. Together, the main sweeps contain 80 regimes. There are 20 random seeds per regime and 50 at the baseline (1630 runs for each learning model).

Before running the main simulations, I calibrated the neural networks' learning rates and numbers of training updates separately for principals and the broker. I considered ranking performance on randomly sampled candidates, stability over time and across random seeds, and computational time.

To show that the findings do not depend on neural-network learning, I repeat the complete design with ridge regression. In the ridge specification, principals use candidate characteristics, while the broker uses both parties' characteristics and interactions between them. I calibrated the ridge penalties at the baseline, considering ranking performance on randomly sampled candidates, stability over time and across random seeds, and prediction error. Three additional ridge experiments, conducted at the baseline and across the $\rho \times \delta$ grid, restrict the broker by capping its fitting sample at the median sample size among principals, retaining only one party's characteristics from each match, or omitting interactions between the two parties' characteristics.

To examine the value and structural consequences of assessment and access, I compare three versions of the broker's service, all using neural-network learning. Full service combines the broker's assessments with access to its contacts and current clients. Assessment-only uses the broker's assessments to rank candidates available through ordinary self-search. Access-only ranks the broker's broader candidate set using principals' own assessments. In each version, the broker still organizes matches, charges the same fee, and observes and learns from the matches it arranges. The service remains fixed

throughout each simulation. Changing the service changes matches, outsourcing, network formation, and subsequent learning. The comparisons capture these combined effects.

Ensemble means are averages across random seeds. The late window covers periods 481–500; the early window covers periods 51–70. Betweenness, constraint, and effective size are computed every 20 periods; their early and late summaries use network snapshots at periods 60 and 500, respectively. Other measures are averaged within each window. Unless otherwise noted in a figure caption, time-series figures begin at $t = 30$ to omit the initialization transient.

I report pointwise 95% Student-t Monte Carlo confidence intervals across independent runs to quantify uncertainty in the estimated means. For comparisons between model versions or regimes, I first calculate outcome differences between runs using the same random seed, then construct intervals for the mean difference across seeds.

Across the main simulations and additional experiments, I assessed how sensitive the reported early and late estimates were to chance variation associated with the particular random seeds used to generate the simulation runs. This assessment covered mean outcomes, paired differences between conditions, and ratios of ensemble output means. I compared estimates based on the first 10 versus all 20 seeds outside the baseline and the first 20 versus all 50 seeds at the baseline.

In the results sections that follow, I describe four salient and sometimes counterintuitive dynamics that emerge from this simple set-up: (i) the broker primarily provides assessment, not access; (ii) access brokerage can raise principals' net output while eroding broker structural advantage; meanwhile, assessment brokerage can sustain brokerage even when no access is provided; (iii) the broker's informational advantage comes from data volume and relational information; and finally, (iv) a broker can be highly central while doing little bridging.

B The Broker Primarily Provides Assessment, Not Access

Most brokered placements involve counterparties that principals could already reach through ordinary self-search. In all 80 regimes, the late-mean share of brokered placements reusing an existing tie exceeds 80%. Yet principals route a majority of their matching requests through the broker in 78 regimes (late means; Figure 1, Panel C). Extensive use of brokerage therefore coexists with relatively little provision of new access. For most placements, the broker assesses and arranges a match between already connected principals.

This assessment role becomes more pronounced over time. At baseline, the outsourcing rate rises from an early mean of 78% to a late mean of 97%, while the access fraction falls from 20% to 11% (Figure 1, Panel A). The early-to-late decline in the access fraction occurs in every regime. The 95% Monte Carlo interval for the decrease excludes zero in each regime (Panel B). Principals thus continue outsourcing as the broker's work becomes increasingly concentrated within existing relationships.

Assessment guides the broker's selection of counterparties: its predicted match values determine how candidates are ranked and which offers are made first. The ranking diagnostic evaluates this component of the service by comparing the broker's and principals' predicted rankings against true rankings by underlying match value on the same randomly sampled candidates. At baseline, late-mean rank correlation is 0.95 for the broker and 0.63 for principals, with a 95% Monte Carlo interval of [0.31, 0.33] for the ranking advantage. That advantage is positive in every regime, with the 95% Monte Carlo interval excluding zero in each regime (Figure 1, Panel D).

The predominance of assessment compared to access also persists when the broker and the principals use ridge regression to make their predictions instead of neural-network learning. In the ridge variant, late-mean outsourcing exceeds one-half in 79 of 80 regimes, and the late-mean share of brokered placements reusing existing ties exceeds 80% in every regime (Figure 1, Panel C). Neural-network learning raises principals' late-mean ranking accuracy in all 80 regimes and narrows the broker's

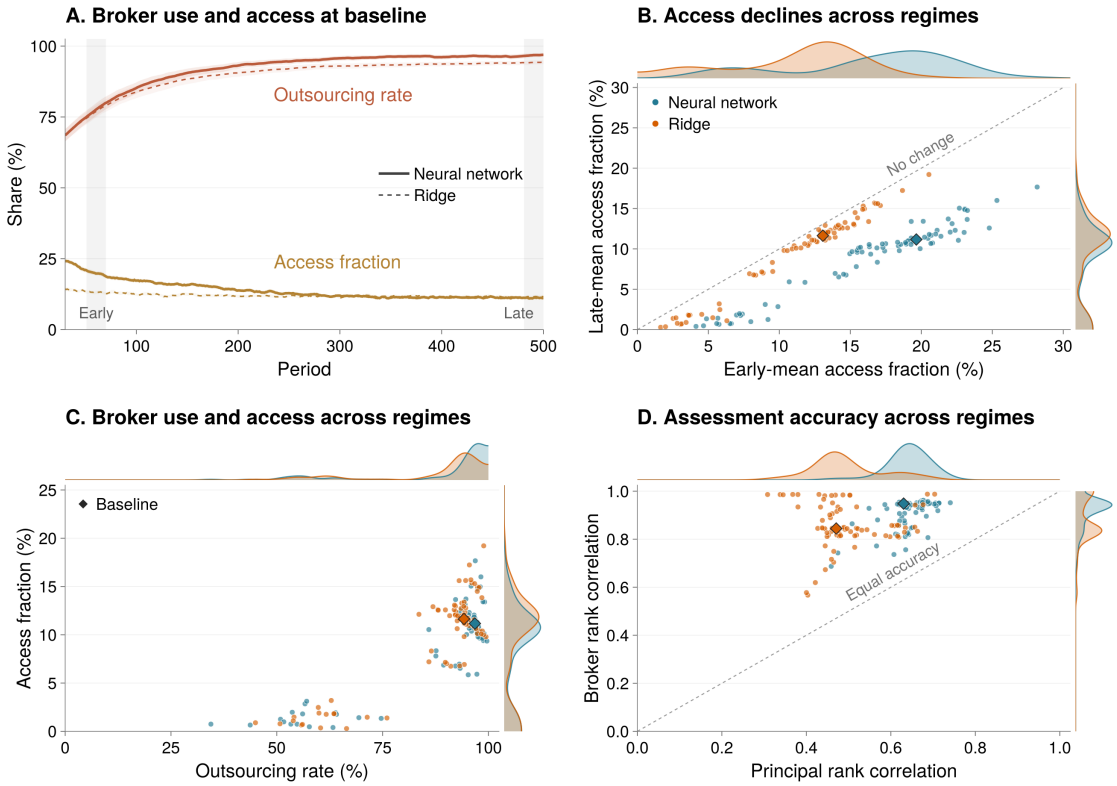


Figure 1: Broker use, access, and assessment accuracy. A: ensemble means after a 5-observation rolling mean within each seed, with pointwise 95% Monte Carlo intervals. B–D: ensemble means for 80 regimes; C–D use late means. Marginal curves show smoothed distributions of regime means, with equal weight per regime and common smoothing across learners.

ranking advantage in 64 (Panel D), yet extensive outsourcing and the predominance of assessment persist.

Together, these findings identify assessment rather than new access as the broker's predominant service. Principals outsource the evaluation and arrangement of matches even when the counterparties involved are already reachable.

C The Value and Structural Consequences of Access and Assessment

The broker ranks candidates more accurately than principals, but ranking accuracy alone does not establish how much its assessments contribute to principals' net output. The assessment and access experiment examines these contributions by comparing full service with assessment-only and access-only service across 11 regimes varying turnover and matching composition. Each simulation uses one version of the broker's service throughout. Changing that service affects which principals match, how much demand they route through the broker, which ties form among principals and with the broker, and what principals and the broker subsequently learn.

C.1 The Broker Can Benefit Principals by Providing Access While Eroding Its Structural Advantage

A broker can increase principals' net output by providing access to counterparties beyond those available through ordinary self-search, even when most of its placements reuse existing ties. At baseline, principals' late-mean net output is 1.02 higher per principal under full service than under assessment-only service, with a 95% Monte Carlo interval of [0.96, 1.08] (Figure 2, Panel A). Principals therefore benefit when the broker provides broader access alongside its assessments, even though only a minority of its placements connect previously unconnected principals.

This benefit depends on how frequently principals need new relationships. When 1% or 3% of participants leave and are replaced each period, principals obtain higher net output under full service than under assessment-only service across all matching compositions. Without turnover, principals obtain less net output under full service than under assessment-only service (Supplementary Figure S4, Panel B). Turnover continually removes principals and their relationships, making the broker's broader contacts useful for finding new counterparties. In stable markets, accumulated relationships and ordinary self-search already provide a viable candidate pool, so the broker does not necessarily improve principals' net output by reaching additional candidates.

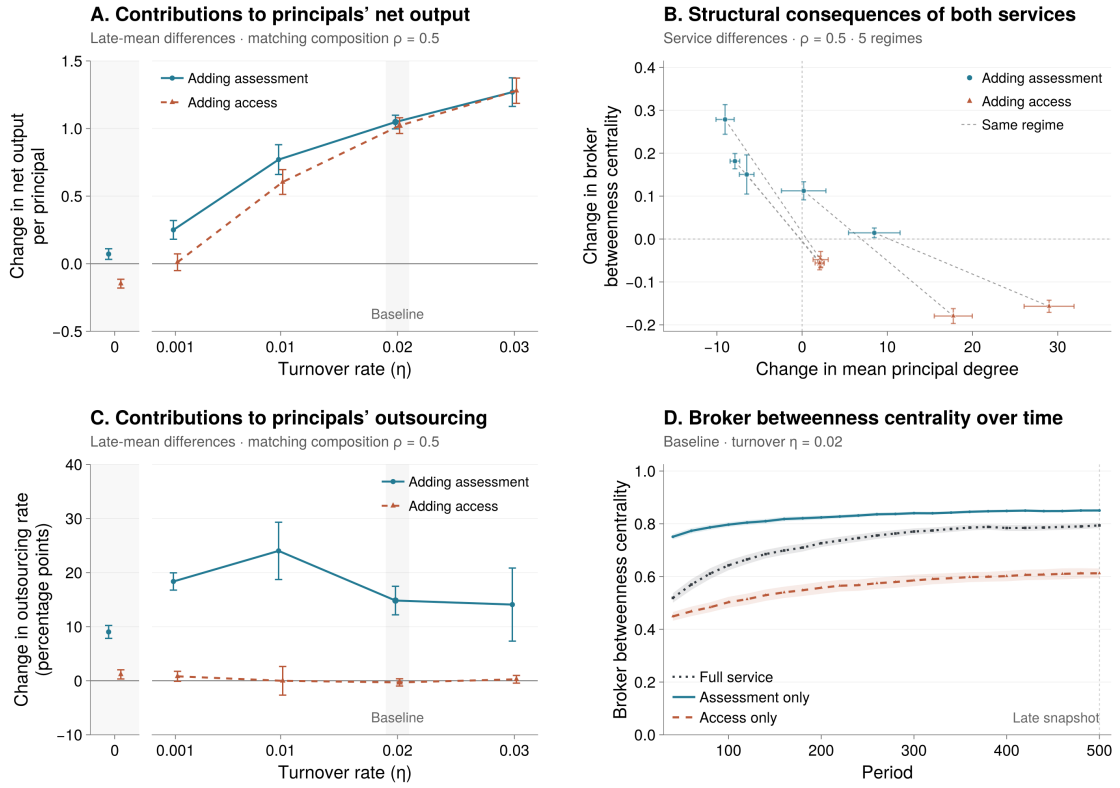


Figure 2: Brokerage with access, assessment, or both. Parameters other than turnover remain at baseline. A–C show differences between services: adding assessment is full service minus access-only; adding access is full service minus assessment-only. Betweenness in B uses period 500; other measures in A–C use late-window averages. Dashed lines in B connect contrasts within each regime; principal degree includes the broker tie when present. Zero turnover is plotted separately in A and C. D shows unsmoothed ensemble means. Bars and ribbons show pointwise 95% Monte Carlo intervals, paired by seed in A–C.

The relative contributions of assessment and access also depend on principals' need for new relationships. Under pure complementarity and no turnover, principals' late-mean net output is 0.26 higher per principal under assessment-only than under access-only service, with a 95% Monte Carlo interval of [0.22, 0.30]. When 3% of participants leave and are replaced each period, principals' late-mean net output is 0.22 higher per principal under access-only than under assessment-only service, with a 95% Monte Carlo interval of [0.13, 0.32] (Supplementary Figure S4, Panels A–B).

Comparing full service with assessment-only service also reveals a structural cost for the broker. At baseline, principals have more network ties on average under full service than under assessment-only service: their late-mean degree is 17.73 rather than 15.52. Degree counts each principal's ties to other principals and its tie to the broker when present. Meanwhile, the broker has lower betweenness centrality at period 500, 0.79 rather than 0.85 (Figure 2, Panel B). Both directions hold in all 11 regimes. The 95% Monte Carlo intervals for both differences exclude zero in every regime.

Providing assessment has the opposite effect on the broker's betweenness centrality. At baseline matching composition ($\rho = 0.5$), the broker has higher betweenness centrality at period 500 under full service than under access-only service at every turnover rate. The 95% Monte Carlo interval for this difference excludes zero at every rate (Panel B).

When the broker arranges a match between previously unconnected principals, they acquire a direct tie. By helping principals form these relationships, the broker erodes the betweenness centrality it would otherwise accumulate. This is a self-liquidating force, not an inevitable absolute decline. At baseline, the broker's betweenness centrality rises even under access-only service, because principals' continued outsourcing still connects them to the broker (Figure 2, Panel D).

C.2 The Broker Can Sustain Demand Through Assessment Without Providing Additional Access

The broker's assessments contribute to principals' net output alongside its broader contacts. At baseline, principals' late-mean net output is 1.05 higher per principal under full service than under access-only service, with a 95% Monte Carlo interval of [1.00, 1.10], compared with a gain of 1.02 over assessment-only service (Figure 2, Panel A).

Principals continue to request brokerage even when the broker evaluates only candidates available through ordinary self-search. At baseline, late-mean outsourcing is 97% under assessment-only service, compared with 97% under full service and 82% under access-only service. Principals outsource more when the broker uses its own assessments rather than theirs in every regime. The 95% Monte Carlo interval for this difference excludes zero in each regime. When the broker adds access to assessment-only service, principals make smaller changes in the share of requests they outsource, increasing it in some regimes and decreasing it in others (Figure 2, Panel C).

These choices depend on principals' satisfaction with brokerage relative to self-search, based on the returns on their past matching requests after costs. Principals can therefore continue using the broker because their satisfaction with its service is high, because their satisfaction with self-search is low, or because both are true. Continued outsourcing establishes the broker's relative attractiveness, while the net-output comparisons establish the contribution of its assessments to principals' returns.

The assessment-only result also connects the ABM to data enclosure. The broker continues to create value for principals when it only assesses candidates they could reach through ordinary self-search. Its knowledge is therefore valuable independently of broader access. The experiment does not model data enclosure itself, because the broker still mediates matches and learns from them, but it establishes its prerequisite: assessment can become a distinct service that could be unbundled from intermediation and sold as predictions or analytics.

D The Broker's Informational Advantage Comes from Data Volume and Relational Information

The broker learns from the matches it arranges across principals, whereas each principal learns only from its own matches. This difference can support an assessment advantage in two ways: the broker has more observations, and it can learn how different parties' characteristics combine to determine the value of a match. Three experiments distinguish these sources by reducing either the broker's training sample or the information it uses about each pairing.

All three experiments use ridge regression for principals and the broker. The reference broker retains data about both parties to each brokered match and learns how their characteristics interact. Each restriction is compared with this reference across 31 regimes varying matching composition and difficulty.

Reducing the broker's training sample sharply narrows its advantage. The first restriction gives it as many observations as a typical principal, sampled from its records of brokered matches, with data about both parties retained. At baseline, its late-mean rank correlation falls from 0.84 in the reference version to 0.76, with a 95% Monte Carlo interval of [0.08, 0.10] for the decrease (Figure 3, Panel A). Its ranking advantage, the broker-minus-principal difference in rank correlation, falls from 0.37 to 0.04, with a 95% Monte Carlo interval of [0.32, 0.35] for the decrease. This larger change also reflects more accurate principal rankings: their late-mean rank correlation rises from 0.47 to 0.72.

Restricting the broker's learning throughout the simulation changes the matches it arranges and thus the observations from which principals learn. Both measures are lower than in the reference version in all 31 regimes. For each measure, the 95% Monte Carlo interval for the decrease excludes zero in every regime. A smaller ranking advantage persists in most regimes. At baseline, the late-mean share of principals' net output from brokered matches falls from 96% in the reference version to 81%, while total net output is 97% of reference output (Panel B). Higher net output from self-search partially offsets lower net output from brokered matches.

Data volume does not explain the advantage by itself. Each principal's observations hold one party fixed: itself. The second broker restriction retains data about only one party to each match, alongside its realized output, while continuing to combine observations across principals. At baseline, the broker's late-mean rank correlation falls from 0.84 to 0.44, and its ranking advantage reverses from 0.37 to -0.25 (Panel A). The 95% Monte Carlo interval for this broker-minus-principal difference is [-0.30, -0.20]. Both measures are lower than in the reference version in all 31 regimes. For each measure, the 95% Monte Carlo interval for the decrease excludes zero in every regime. In 21 regimes, the Monte Carlo interval for its ranking advantage lies entirely below zero. At baseline, the late-mean share of principals' net output from brokered matches falls from 96% in the reference version to 55%, and total net output is 90% of reference output (Panel B).

Retaining data about both parties also differs from learning how they combine. The third restriction omits the interaction term from the broker's predictions: it uses data about both parties, but one party's contribution cannot depend on the other's characteristics. At baseline, the broker's late-mean rank correlation falls from 0.84 to 0.47 and its ranking advantage from 0.37 to -0.18 (Panel A). The 95% Monte Carlo interval for the 0.55 decrease in ranking advantage is [0.51, 0.60]. At baseline, the late-mean share of principals' net output from brokered matches falls from 96% in the reference version to 61%, and total net output is 90% of reference output (Panel B).

The broker's rank correlation is lower than in the reference version in all 30 regimes involving complementarity. The 95% Monte Carlo interval for the decrease excludes zero in every regime. Its ranking advantage also narrows in all 30 regimes. The 95% Monte Carlo interval for this decrease excludes zero in 29. When underlying match value depends only on individual quality, the restriction instead produces a higher broker rank correlation and a larger ranking advantage. Here, underlying

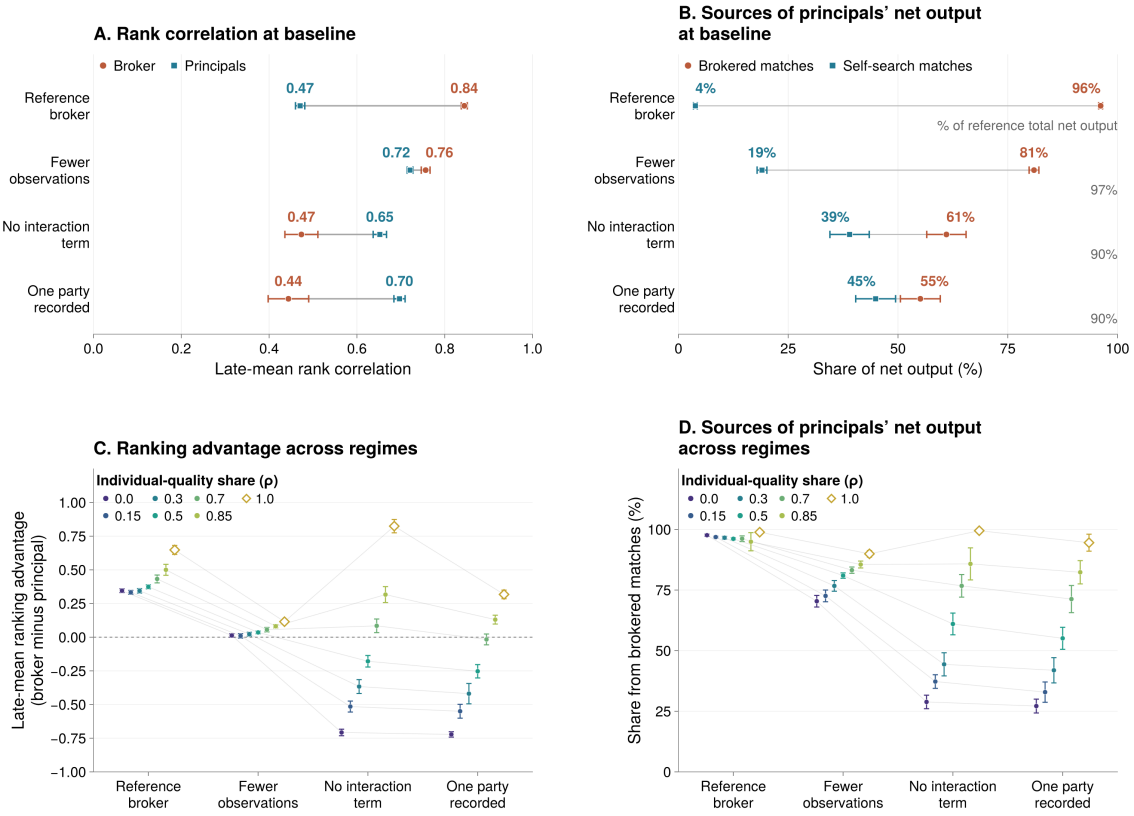


Figure 3: Sources of the broker's advantage. Ensemble late means from the ridge experiments. Rank correlations use the same randomly sampled candidates for principals and the broker. In C–D, parameters other than ρ remain at baseline; gray lines connect the same regime across versions. Whiskers show pointwise 95% Monte Carlo intervals. C uses within-seed broker-minus-principal differences; B and D average seed-level net-output shares. B's right-hand percentages compare ensemble-mean total net output with the reference version.

match value depends on the parties' separate contributions, rather than how their characteristics interact. Omitting the interaction term matches this simpler relationship and avoids estimating interactions that do not contribute to underlying match value.

These effects illustrate the importance of relational assessment. Removing data about one party or the interaction term reduces the broker's ranking advantage most under pure complementarity (Panel C; Supplementary Figure S7, Panels A–E). At each difficulty level, these restrictions also shift a larger share of principals' net output from brokered matches toward self-search as complementarity contributes more to underlying match value (Panel D; Supplementary Figure S7, Panels F–J). Learning how particular parties fit together, beyond assessing their individual quality, improves the broker's rankings and increases the share of principals' net output that comes from brokered matches.

Repeated vicarious relational work thus supplies both a larger sample and knowledge of which pairings succeed. Retaining data about both parties and learning how their characteristics combine make these observations useful for relational assessment.

E A Broker Can Be Highly Central While Doing Little Bridging

The broker can become more central even as a smaller share of its placements connects previously unconnected principals. At baseline in the main simulations, broker betweenness centrality rises from 0.57 at period 60 to 0.79 at period 500, while the access fraction falls from 20% to 11%. A larger share of the shortest paths between principals therefore passes through the broker, even as its work becomes increasingly assessment-based. Such a central position is usually associated with connecting otherwise disconnected parties, yet a growing share of the broker's work takes place within relationships that already exist.

The broker's betweenness centrality depends on two kinds of ties: those connecting principals to the broker and those connecting principals to one another. Continued outsourcing connects principals to the broker. When the broker arranges matches within existing relationships, it sustains these connections without creating additional direct ties among principals. Providing access, by contrast, creates direct ties that offer alternative shortest paths bypassing the broker. These opposing effects appear in the assessment and access experiment: providing access lowers broker betweenness centrality, whereas providing assessment raises it (Figure 2, Panel B).

Although brokerage is predominantly assessment-based across matching compositions, the broker's work becomes even more concentrated on assessment and its position in the network becomes more central as underlying match value shifts from depending mainly on individual quality to depending entirely on complementarity. At baseline turnover and difficulty, moving from $\rho = 0.85$ to $\rho = 0$ raises the late-mean share of brokered placements within existing relationships from 86% to 90% and broker betweenness centrality from 0.52 to 0.86 (Figure 4, Panel A). Both differences occur at all 5 turnover rates. The 95% Monte Carlo intervals for both differences exclude zero at every rate.

Extensive outsourcing and predominantly assessment-based brokerage do not by themselves ensure high broker betweenness centrality. The distribution of ties among principals also matters. When underlying match value depends only on individual quality ($\rho = 1$), the baseline late-mean share of brokered placements within existing relationships is 89% and outsourcing remains high at 97%, but broker betweenness centrality is only 0.15. The same counterparties are attractive to many principals, allowing relationships to concentrate around them. The late mean of the maximum principal degree is 985.19, compared with 136.36 when underlying match value depends entirely on complementarity. Mean principal degree differs much less, at 19.36 and 15.77, respectively (Figure 4, Panel B). A modest average degree can therefore conceal highly connected principals that provide alternative shortest paths that bypass the broker.

The persistence of principals' relationships also matters. Turnover continually removes principals and their ties, limiting the accumulation of paths that bypass the broker. At baseline matching composition and difficulty, replacing 3% of principals each period rather than none lowers late-mean principal degree from 77.35 to 16.33 and raises broker betweenness centrality from 0.12 to 0.82 (Figure 4, Panel C). The access fraction also rises, from 1% to 15% (Panel A). A larger access fraction therefore does not necessarily imply lower broker betweenness centrality: newly formed ties can be offset by the loss of existing relationships.

The data and learning capabilities behind the broker's assessment advantage also help sustain its bridging position. At baseline in the ridge experiment, omitting the interaction term reduces late-mean outsourcing from 94% to 57% and broker betweenness centrality from 0.75 to 0.16 (Figure 4, Panel D). The late-mean access fraction remains low, at 10% compared with 12% for the reference broker. In both versions, most brokered placements reuse existing relationships, but the broker has much lower betweenness centrality when principals use its service less. Learning how parties fit together thus helps sustain principals' use of the broker and the connections that make it central.

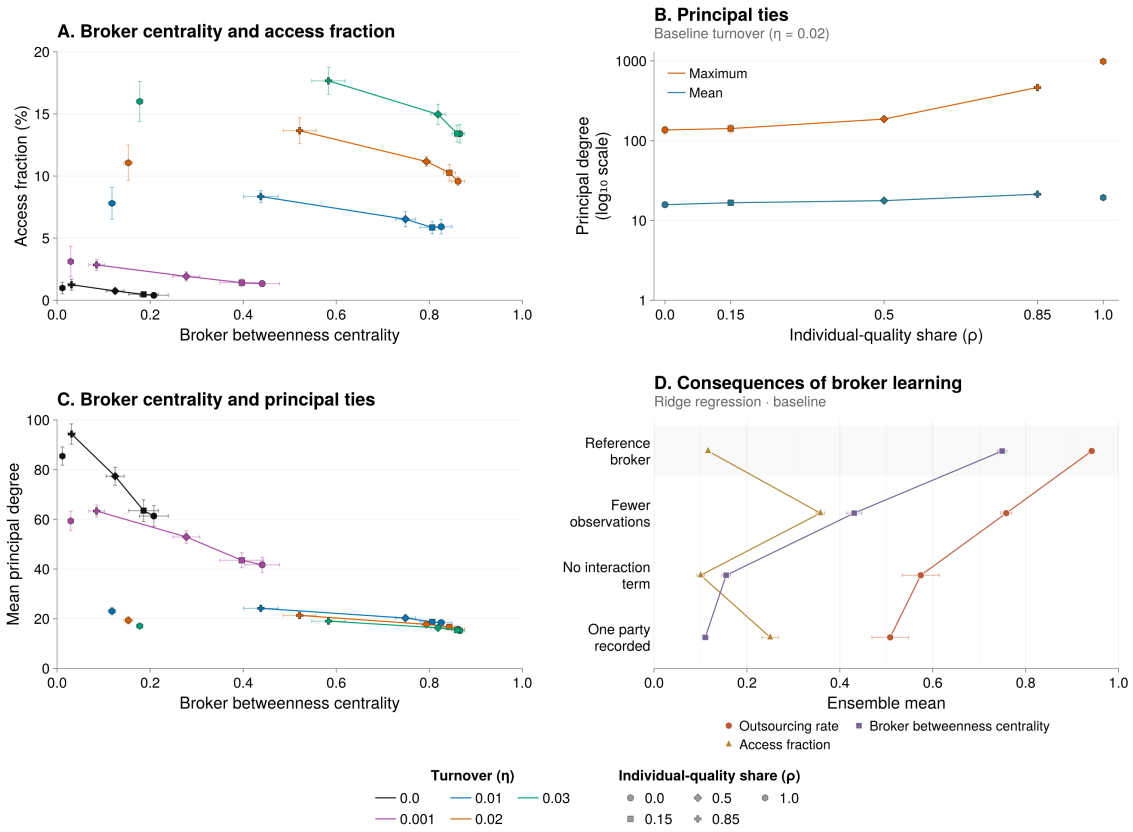


Figure 4: Bridging position and bridging behavior. Points show ensemble means, using betweenness at period 500 and late-window averages for other measures; whiskers show pointwise 95% Monte Carlo intervals. A–C use neural-network learning; parameters not varied in each panel remain at baseline. Principal degree includes the broker tie when present; maximum degree is calculated within each period before averaging. Lines connect matching compositions at fixed turnover in A and C and serve as visual guides across learning restrictions in D.

To assess how the structural findings depend on the measure used, I repeat the comparisons using Burt's constraint and effective size. Supplementary Figure S5 covers matching composition and turnover, and Supplementary Figure S6 covers service provision and learning. When underlying match value depends only on individual quality and turnover is 3%, providing access increases effective size, despite reducing betweenness centrality (Panel B).

Repeated vicarious relational work generates knowledge that can sustain principals' use of the broker. Sustained use converts the value of assessment into structural centrality, while the concentration and persistence of principals' own relationships shape how much centrality the broker attains. High broker betweenness centrality can therefore coexist with little bridging.

F Propositions

The simulation results and theoretical argument motivate the following falsifiable propositions for future empirical research:

P1. ***When complementarity matters, assessment is the primary source of brokerage value.*** In matching markets that reward complementarity, brokers add value primarily by assessing counterparties rather than by bridging disconnected parties. The value of this assessment role increases as complementarity contributes more to underlying match value.

P2. ***Relationship disruption increases the value of access brokerage.*** Clients gain more from brokers' ability to find new counterparties when market participants frequently leave and are replaced than when they can rely on stable relationships.

P3. ***When the broker records and accumulates data about its matching work, brokerage widens informational asymmetry.*** Repeated brokerage increases the broker's assessment advantage over its clients, even though those clients also learn from the matches it arranges.

P4. ***Assessment brokerage produces high broker centrality with little bridging behavior.*** Sustained outsourcing converts the value of assessment into greater structural centrality for the broker. Brokers whose assessments remain valuable relative to clients' independent alternatives become highly central even when most of their matches involve counterparties already connected to one another.

P5. ***Assessment enables the broker to become a principal.*** Corporate brokers whose assessments create value independently of privileged access are more likely to use that capability on their own behalf, either to acquire and sell resources they previously intermediated (resource capture) or to sell predictions or analytics directly (data enclosure).

V Transient Brokerage and Its Conditions of Possibility

The model formalizes how brokers accumulate and sustain informational and structural advantages, but it stops before the transition to principals. I now return to the theoretical argument to explain how a corporate broker may convert those accumulated advantages into resource capture or data enclosure.

Outsourced relational work performed at scale by corporate brokers creates the conditions under which this conversion can occur. The broker dismantles the relational package previously negotiated with its clients and replaces it with a new configuration organized around the broker's ownership of the resources it used to intermediate (*resource capture*), or the monetization of the information generated through prior matching activity (*data enclosure*).

These configurations are two qualitatively different forms of capture, which can be pursued in parallel and along with standard brokering. In *resource capture*, the broker takes inventory risk and starts acting as a principal in the market it used to intermediate. Counterparties interact only with the broker, without creating direct ties with one another. As the broker expands its principal activity and shrinks its matching services, former clients become counterparties or competitors. In *data enclosure*, the broker monetizes information directly by selling predictions, analytics, or other information services, often as a subscription.

The assessment-only experiment provides a formal precursor to this transition. In that experiment, the broker provides no access beyond ordinary self-search, yet its assessments remain valuable to principals and sustain outsourcing. Data enclosure takes this separation one step further: the broker monetizes its accumulated assessment capability without necessarily continuing to mediate the transactions it evaluates.

What makes capture possible, I suggest, is primarily the broker's informational advantage, not its structural position. Structural and informational advantages are decoupled, and capture may occur even if the broker's centrality is declining. This is because constructing viable relational packages

generates an informational byproduct that the broker can leverage. Two features of the broker's informational landscape facilitate capture: cumulative knowledge of complementarities coming from the *outsourced* nature of its relational work, and information infrastructures arising from its *corporate* nature. Other actors cannot easily replicate either advantage.

(i) The role of complementarity information. The broker's vicarious work generates knowledge of how different parties pair together, what kinds of matches succeed across the market, and what relational packages support different transactions. For regular market actors, this complementarity information is hard to replicate. Unlike attributional information about the individual quality of counterparties, which any actor with sufficient market exposure can in principle acquire, complementarity information is generated through the repeated observation of various pairings. The broker holds it because it does the matching for a variety of clients; clients, who experience only their own matches, do not—unless they purchase or otherwise acquire it. Complementarity information supports resource capture, because it allows the broker to identify which resources or counterparties are most likely to generate profitable matches, so that it can capture them strategically and offload them gainfully. Complementarity information also supports data enclosure because it can be packaged into prediction or analytics products and services.

(ii) The role of information infrastructures. The corporate broker's information infrastructure explains *how* its informational advantage can be preserved and converted. The taxonomies, classification systems, contractual templates, and algorithmic sorting procedures that the broker develops to perform vicarious work at scale and in a distributed manner, through the everyday work of its employees and representatives, are also the infrastructures through which capture can be realized. They can be repurposed to route resources to the broker itself, to put these resources back in circulation profitably, or to organize data into a marketable product. Moreover, the corporate form gives the broker the legal privileges required to build, own, and protect these infrastructures over time (Coleman 1982; Pistor 2019).

These features, and transient brokerage itself, can be found across markets. I illustrate each configuration with an example.

Dealers in collectible markets. Collectors of art, fine wine, rare books, or similar specialty goods may seek trades through broker-dealers who know the market. Each collector has distinct tastes and holdings; match quality depends on multidimensional complementarity between what one party has and what another wants. Dealers accumulate knowledge of collector preferences across transactions, and they may decide to leverage it and start holding inventory, for instance, by launching and operating a gallery.

Interdealer brokerage in over-the-counter (OTC) financial markets. Interdealer brokers (IDBs) are specialized intermediaries in OTC financial markets (e.g., bond markets, foreign exchange, commodities). IDBs help institutions like banks and dealers conclude large, complicated, or anonymous transactions in the absence of a centralized trading place. They solve a hard matching problem: the right counterparty for a trade depends on inventory positions and risk appetites that parties wish to keep secret. IDBs must know who wants what, negotiate prices across hidden order interest, preserve anonymity until execution, and manage long-term client relationships to remain in the order flow. The complementarity information they accumulate, about who will absorb what positions at what price under what conditions, is substantial. Resource capture, however, faces regulatory and reputational hurdles. Instead, IDBs have monetized accumulated information through data divisions that package historical OTC pricing and other data products and sell them to the same dealers whose trades generated the data and to a wider set of buy-side firms and regulators (see, e.g. Parameta Solutions 2026; TP ICAP 2026).

In summary, three elements enable capture or enclosure: a matching problem that makes complementarity information valuable, an informational advantage accumulated through repeated vicarious

relational work, and an institutional architecture that allows that information to be aggregated, organized, and protected.

VI Discussion

A Structural Advantage as a Product of Relational Work

Within the agent-based model, the broker's centrality is a product of its relational work. The broker is central because many principals outsource to it repeatedly: good assessments help keep brokerage attractive relative to self-search. Its centrality rises and falls with outsourcing and satisfaction.

The simulations show that the broker's relational work endogenously produces its structural advantage. This process generates two patterns that existing theory does not readily predict. The first is that the broker tends to be most central exactly where it bridges least in practice. Across variation in matching composition at a fixed turnover rate, centrality is negatively associated with the access fraction (Figure 4, Panel A): in the complementarity-dominated markets where the broker is most prominent, almost all its matches re-pair principals already tied to one another. A strong bridging position coincides with weak bridging behavior. Occupying a strategic position and using it for one's work are different things that theory should not conflate.

The second pattern concerns the sustainability of brokerage. A long-standing expectation is that brokerage tends to undermine its own structural basis (Burt 1992, 2002; Stovel, Golub, and Milgrom 2011). Each successful introduction creates a direct tie that can reduce future dependence on the broker. Access therefore creates a self-liquidating force, not an inevitable decline in broker centrality.

The experiment makes this distinction clear. At baseline turnover, the broker has lower betweenness centrality under full service than under assessment-only service, yet all three broker models gain centrality over the observed period (Figure 2, Panels B and D). Providing access can thus lower centrality relative to assessment-only without causing centrality to decline over time. Turnover removes ties and renews opportunities for brokerage, while valuable assessments can sustain outsourcing. A broker can therefore become more central even while its introductions weaken one source of its structural advantage.

The results also suggest a precise mechanism for the decoupling of structural and informational advantage posited in Section V. The two advantages respond to different features of the matching problem. The broker's structural prominence tracks matching composition, rising as matching becomes more complementarity-driven, while its ranking advantage responds to the difficulty of learning the complementarity component, which can vary independently. A broker can be highly central despite a modest ranking advantage or hold a decisive ranking advantage from an unremarkable position. Betweenness centrality is best understood not as a static measure of opportunity but as a product of the relational work the broker performs.

B Scope Conditions and Generalizability

The concept of transient brokerage applies most directly to matching markets in which (i) matching is complex enough that the value of a match depends not only on the individual quality of each party but also on their complementarity and (ii) a corporate broker operates as a repeat player accumulating information across transactions. The further a case departs from these conditions, the weaker the predictions of the theory. When the relational dimension is weak, the broker's advantage rests primarily on data volume about multiple counterparties, making its informational position less privileged. When the broker is a natural person, they lack the temporal persistence, infrastructural capacity, and legal architecture that the corporate broker relies on.

The dynamics of transient brokerage also depend on the nature of the brokered resource and the institutional environment. In a goods market where items are bulky and physical delivery is required, the costs and logistics of capture differ substantially from those in a market for services or a market where goods are virtual or ownership passes without physical possession. In some markets, regulation may slow or prevent capture. Market structure matters as well. The model considered in this article features a single broker. With multiple brokers competing for the same clients or matches, the cross-market data pool is fragmented, and each broker may accumulate a smaller informational advantage, unless one or a few brokers dominate the market. A systematic treatment of broker competition is deferred to future work.

Transient brokerage is relevant to a particular configuration of brokerage, in which the matching problem, corporate form, and information accumulation give rise to a distinctive trajectory. The sociological literature has used brokerage to describe a wide range of phenomena, from immigrant children translating for their parents to marriage matchmakers and political brokers. The relational-work view of brokerage developed in this article can be applied to what many of these so-called brokers do. Without the repeat-player structure of corporate brokerage, the legal and infrastructural architecture of the firm, and an economic matching problem complex enough to sustain an informational advantage, however, the pathway to capture is not the same.

These scope conditions point to candidate domains in which to test the theoretical propositions. Real estate and various types of financial intermediation, including interdealer brokerage, combine complex matching, corporate repeat-player brokers, and meaningful complementarity information. Import-export management and data markets also share many of these features.

C The Political Economy of Corporate Intermediation

The corporate broker is a vehicle for the accumulation of information generated by the interactions between others, and the principals whose matching activity produced the information do not participate in the profits. The asymmetries that make capture possible are not just a byproduct of a broker's size or expertise. They are legally constituted. Corporate brokers assemble cross-market data and build classification systems and information infrastructures under the legal protections of the corporate form, which allows vicarious relational work to solidify into a durable informational moat. Whether a broker can convert its informational position into resource ownership or data provision depends on the institutional environment as much as on network position.

When the broker becomes a principal, it changes who gets connected to whom. The principals observe far fewer matches than they otherwise would, and most matches in the market teach them nothing at all. The informational asymmetry that makes capture possible is widened by capture itself: the broker takes control not only of the resource but of the data its placement generates. Transient brokerage shifts how value is distributed, channeling the returns of participants' collective relational activity to the corporations that record and mine the data it generates.

Capture by corporate brokers has become more common since the 1980s alongside three transformations that have reshaped market intermediation: the turn to finance, which expanded the share of economic activity routed through specialized intermediaries (Carruthers 2015; Krippner 2011); the digital turn, which dramatically lowered the cost of accumulating, storing, and analyzing the data that brokerage generates while also expanding the demand for predictive analytics (Cohen 2019; Gandy 1993); and the rise of platform capitalism, which produced multi-sided market operators that own the relational infrastructure of matching while monetizing the data it generates (Sadowski 2020; Viljoen, Goldenfein, and McGuigan 2021). Transient brokerage is one of the mechanisms that link these transformations together and contributes to a broader shift in how value is extracted from economic activity.

Significant evidence from financial markets shows that corporate brokers that once facilitated matches between others have increasingly become principals, either by capturing the activity they used to intermediate or by monetizing data. The demutualization of stock exchanges is a prime example. As late as the mid-1990s, almost all major global exchanges operated as member-owned mutual organizations. By the mid-2000s, a majority had introduced electronic trading systems that allowed them to grant trading access directly to investors, bypassing the services of member firms, and had converted to for-profit shareholder structures (Aggarwal 2002; Petry 2021). For-profit exchanges now derive a growing share of revenue from proprietary data feeds and trading technology (MacKenzie 2021; Pardo-Guerra 2019).

A parallel transformation has taken place in interdealer brokerage for over-the-counter (OTC) markets, where the data divisions of prominent brokers now generate substantially higher operating margins than the broking activities themselves; this is the case, for instance, of the Parameta Solutions data division of TP ICAP, the largest interdealer broker (TP ICAP 2026). In consumer markets, Visa offers another example. Reorganized in 1970 as a member-owned cooperative of issuing banks, Visa operated for decades as a non-profit utility that brokered payment authorizations between cardholders, merchants, and banks (Stearns 2011). Following its 2007 restructuring and 2008 IPO, Visa converted into a for-profit corporation that now derives substantial revenue from data, analytics, and risk solutions services, alongside the network and processing fees generated by its core authorization and clearing activities (Visa Inc. 2024).

VII Conclusion

This article reconceptualizes brokerage as outsourced relational work and explains how repeated matching can produce an informational advantage that a corporate broker may later convert into resource capture or data enclosure. The formal illustration demonstrates one possible pathway through which repeated vicarious relational work can produce informational and structural advantages, and it generates propositions about when those advantages should arise. By pooling observations across clients, the broker learns how both parties' characteristics combine and develops assessments that are particularly valuable when the value of a match depends on complementarity. Those assessments can sustain outsourcing even when principals can reach counterparties themselves. Continued outsourcing connects principals to the broker and can increase its structural prominence. When the broker arranges matches between previously unconnected principals, it also creates direct ties that can weaken that position. The broker can therefore be highly central while doing little bridging.

The very conditions that make brokerage possible and valuable, such as the scarcity of information and the complexity of the matching problem, create opportunities for the broker—particularly the corporate broker—to take control of resources. At the transition point, the broker dismantles the relational package negotiated with its principals and collapses the boundary between *vicarious* and *personal* relational work to pursue its own advantage. It abandons or reconfigures the service-tie configuration that existed and (re)negotiates a new one, premised on the broker's direct ownership and control of formerly brokered resources or the data they generated. In the latter case, the accumulated data can also be mined internally or licensed out to feed proprietary behavioral prediction, classification, and scoring products (Fourcade and Healy 2017, 2024) for use in markets distinct from those in which they originated (Viljoen 2021; Zuboff 2019).

This article helps explain why the digital turn has facilitated broker capture at scale. Because information is central to brokerage, changes in the information ecosystem reshape the conditions under which transient brokerage becomes available, and digital technology has changed those conditions decisively. Relational infrastructures that once consisted of internal handbooks, contractual templates, and bureaucratic procedures now incorporate data taxonomies, algorithmic sorting, and interface designs that constitute the relational packages they administer in real time and at scale. Digital

technology amplifies these asymmetries and shortens the path from accumulated information to ownership or monetization.

Digital platforms and algorithmic marketplaces partly automate the work of matching through auction protocols, ranking algorithms, and rating systems. Where platform operators perform this work on behalf of others, automation reorganizes vicarious relational work rather than eliminating it. The operator administers categories, ratings, and defaults that shape how users interact and transact. It also accumulates information across users. It can leverage this information to acquire and sell resources on its own account (resource capture), or to sell predictions and analytics directly (data enclosure). Both activities can coexist with continued brokering. Brokerage becomes transient when matching on behalf of others becomes secondary or ceases.

For example, Amazon operates as both a marketplace for third-party sellers and a retailer selling on its own account, showing how these roles can coexist. Zhu and Liu (2018) found that Amazon was significantly more likely to begin directly selling products with higher sales and better customer reviews, using information available to it as the marketplace operator. This illustrates how information accumulated through intermediation can guide the operator's own transactions, including competition with the sellers it serves. Brokering continues alongside direct selling. The broker can strengthen its advantage while becoming a competitor to the clients whose transactions helped produce it.

Two main directions warrant future work. On the modeling side, the ABM developed here has many interacting components and carries the usual caveats of this kind of hand-crafted simulation: its results illustrate the theory rather than establish it. The resource capture and data enclosure mechanisms are introduced theoretically but not yet formalized. On the theoretical side, the framework does not yet fully address how platforms and algorithmic marketplaces automate the vicarious relational work that brokers historically performed. Both directions point toward the same underlying question about the impact of the digital turn on brokerage, the institutions that rely on it, and the actors that perform it or benefit from it.

The digital turn has brought about a series of transformations. The informational byproduct of relational matching has come to rival the value of the matches themselves. Relational work is increasingly valued for the data it generates and performed by machines rather than people. The informational asymmetry between brokers and their principals has widened. Existing relational packages are being dismantled to extract value from the information and infrastructures associated with them. These shifts redraw the boundaries on which brokerage rests: between internalized and outsourced relational work, and between vicarious and personal relational work. Brokerage becomes a transient position, held so long as brokering remains more profitable than resource capture.

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Appendix A
Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of principals, with principal set $[N] = \{1, \ldots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \ldots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Principal type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction update weight (exponentially weighted moving average, EWMA).
L	learning function	Neural network (NN, default) or ridge regression.
S_b	broker service	Full (default), assessment-only, or access-only.
λ_a, λ_b	positive scalars	Principal and broker ridge slope penalties; calibrated experiment values are $\lambda_a = 0.001$ and $\lambda_b = 0.001$.
a_b	broker ridge variant	Pair, size-matched, single-principal, or additive.
$E_{\text{init},a}, E_{\text{init},b}$	scalars	Principal and broker initial training steps; both are 100 in the reporting simulations.
$E_{\text{steps},a}, E_{\text{steps},b}$	scalars	Principal and broker minimum training steps per fitting call; both are 50 in the reporting simulations.
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	scalars	Principal and broker Adam learning rates; reporting values are 0.003 and 0.03, respectively.
W_p	scalar	Training-window horizon, in periods.
M_{train}	scalar	Per-call training observation cap.

Symbol	Dimension/domain	Definition
Principal types, networks, and output		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.
$X^t = [x_1^t, \ldots, x_N^t]$	$\mathbb{R}^{d \times N}$	Principal type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Principal type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized match output for pair (i,j).
Match-output function		
Q	scalar	Additive offset in underlying match value.
ρ	scalar	Controls the relative weighting of individual quality and complementarity: $\rho = 0$ gives pure complementarity; $\rho = 1$ gives pure individual quality.
δ	scalar	Strength of the pair boundary in match-specific complementarity.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Common ideal principal type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite complementarity matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric boundary matrix.
Calibration and costs		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation threshold.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
d_B	integer	Broker symmetric pair-feature dimension, $d + d(d+1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_B}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Principal i history; observation ℓ is partner principal type $\Xi_{i,\cdot\ell}^t$ and realized match output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and realized match output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Principal i NN weights or ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i,j)$	scalar	Broker's predicted match value for unordered pair $\{i,j\}$.
$a_{i \to j}^t$	scalar	Principal i's assessment of candidate j.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Principal i's mean realized match output by partner.
$C_{i\cdot}^t$	vector	Principal i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that principal i has ever used the broker.

Symbol	Dimension/domain	Definition
M_i^t	set	Principal i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \begin{pmatrix} \vartheta, \mathsf{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \\ \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \\ \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \end{pmatrix}.$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

The fixed calibration moments are calculated across all unordered pairs of distinct principals within each of 50 independently generated calibration markets of 1,000 principals, then averaged with equal weight across markets. Define

$$U_{ij} = \frac{x_i^\top c + x_j^\top c}{2}, \qquad V_{ij} = x_i^\top A x_j, \qquad H_{ij} = \text{sign}(x_i^\top B x_j) V_{ij},$$

$$\widetilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}, \qquad \widetilde{W}_{ij}(\delta) = \frac{V_{ij} + \delta H_{ij} - (\mu_V + \delta \mu_H)}{\sqrt{\sigma_V^2 + 2\delta \,\text{Cov}(V, H) + \delta^2 \sigma_H^2}}.$$

With $\kappa(\delta)$ denoting the average within-market covariance between the standardized individual-quality and complementarity components in the calibration markets, the underlying match value excluding Q is

$$f(x_i, x_j) = \frac{\sqrt{\rho}\,\widetilde{U}_{ij} + \sqrt{1-\rho}\,\widetilde{W}_{ij}(\delta)}{\sqrt{1 + 2\sqrt{\rho(1-\rho)}\,\kappa(\delta)}}.$$

The normalization coefficients depend on ρ and δ, are shared across simulations, and remain fixed through turnover. For a given pair, underlying match value is $Q + f(x_i, x_j)$. A completed match produces realized match output

$$q_{ij} = Q + f(x_i, x_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.
Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T \times m}$.

$S^0 \leftarrow \textsc{Initialize}(\vartheta, \text{seed})$
for $t = 1, \ldots, T$ **do**
    $(S^t, \mathcal{R}^t, Y^t) \leftarrow \textsc{PeriodUpdate}(S^{t-1}, t, \vartheta)$
    $\mathcal{M}_{t\cdot} \leftarrow Y^t$
return $(S^T, \mathcal{M})$

3 Learning and Prediction

Let $\mathsf{L} \in \{\text{NN}, \text{Ridge}\}$ select the learning function. The default is $\mathsf{L} = \text{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their original scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\text{NN}_\Theta(\xi) = w_2^\top \,\text{ReLU}(W_1\xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_B = d + \frac{d(d+1)}{2}, \qquad \psi_b(x_i, x_j) = \left[x_i + x_j;\ \text{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2} \right) \right].$$

Principal networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_B, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
L	learning function	NN or Ridge.
a_b	broker ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or ridge parameters (α, β).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	Neural network weights and biases.
d_B	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch mean squared error (MSE) objective.
λ	positive scalar	Learner-specific ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the principal and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants (0.9, 0.999, 10^{-8}).
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.

Local variables for training (continued)

Symbol	Dimension/domain	Definition
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M_{train}.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Principal and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d\times\lvert L_i\rvert}$, $\mathbb{R}^{\lvert L_i\rvert}$	Principal i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{\lvert L_b\rvert}$	Broker fitting view using the selected input map.
r_ℓ	$\{1,2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}}\times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}, persistent Adam state (m, v, τ).

for $e = 1, \ldots, E$ **do**
  $g \leftarrow \nabla_\Theta \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1}\sum_{\ell=1}^n \left(\text{NN}_\Theta(\Xi_{\text{tr},\cdot\ell}) - y_{\text{tr},\ell}\right)^2$
  $\tau \leftarrow \tau + 1$
  $m \leftarrow \beta_1 m + (1-\beta_1)g, \quad v \leftarrow \beta_2 v + (1-\beta_2)\, g \odot g$
  $\Theta \leftarrow \Theta - \eta_{\text{lr}} \dfrac{m/(1-\beta_1^\tau)}{\sqrt{v/(1-\beta_2^\tau)} + \epsilon}$
return $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function L, learner $c \in \{a, b\}$, parameters Θ, raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}}\times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific ridge penalty λ_c, and fitting parameters.

if $\mathsf{L} = \text{NN}$ **then**
  $E \leftarrow \max\{E_{\text{steps},c}, \lceil E_{\text{init},c}\Delta/n_{\text{total}}\rceil\}$.
  $(\Theta, \mathcal{O}) \leftarrow \textsc{TrainNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr},c}, \mathcal{O})$.
else
  $(\alpha, \beta) \leftarrow \arg\min_{a,b} \left\{ n^{-1}\sum_{\ell=1}^n (a + b^\top \Xi_{\text{tr},\cdot\ell} - y_{\text{tr},\ell})^2 + \lambda_c \|b\|_2^2 \right\}$.
  $\Theta \leftarrow (\alpha, \beta)$.
return Θ and, for NN, $\mathcal{O}$.

The principal input is always the prospective partner type. The broker ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair}, \text{size-matched}\}, \\ x_{r_\ell,\ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median principal sample size among principals with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\textsc{PredictPrincipal}(i, x_j) = \begin{cases} \mathrm{NN}_{\Theta_i}(x_j), & \mathsf{L} = \mathrm{NN}, \\ \alpha_i + \beta_i^\top x_j, & \mathsf{L} = \mathrm{Ridge}, \end{cases}$$

and

$$\textsc{PredictBroker}(i, j) = \begin{cases} \mathrm{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathsf{L} = \mathrm{NN}, \\ \alpha_b + \gamma_b^\top \psi_b(x_i, x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b \in \{\text{pair}, \text{size-matched}\}, \\ \alpha_b + \gamma_b^\top (x_i + x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b,\mathrm{tr}}, & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^\top x$. Before any ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b,\mathrm{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S, period t, learning function L, and fitting parameters.

for $i = 1, \ldots, N$ in parallel **do**
  $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
  if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
    Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \le W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \ldots, n_i\}$. If $|L_i| > M_{\mathrm{train}}$, keep M_{train} spaced indices across L_i, including its first and last indices (or only its last index when $M_{\mathrm{train}} = 1$).
    $\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.
    $\Theta_i \leftarrow \textsc{FitPredictor}(\mathsf{L}, a, \Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.
$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
if $n_b > 0$ and $\Delta_b > 0$ **then**
  Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \le W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \ldots, n_b\}$. If $|L_b| > M_{\mathrm{train}}$, keep M_{train} spaced indices across L_b, including its first and last indices (or only its last index when $M_{\mathrm{train}} = 1$).
  If $\mathsf{L} = \mathrm{Ridge}$ and $a_b =$ size-matched, replace L_b by the uniform size-matched sample defined above.
  Construct $Z_{b,\mathrm{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\mathrm{tr}} \leftarrow (y_b^t)_{L_b}$.
  $\Theta_b \leftarrow \textsc{FitPredictor}(\mathsf{L}, b, \Theta_b, Z_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\mathrm{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$
$$r = f_r\, \bar{q}_{\mathrm{cal}}, \qquad \phi = c_s = \lambda_c\, \bar{q}_{\mathrm{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
f_r	scalar	Reservation coefficient; sets $r = f_r\,\bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d\times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d\times d}$	Empirical principal type second moment.
B_0	$\mathbb{R}^{d\times d}$	Raw zero-trace symmetric boundary matrix.
B_{raw}	$\mathbb{R}^{d\times d}$	Orthogonalized pre-normalization boundary matrix.
$Z_{b,\text{seed}}, y_{b,\text{seed}}$	matrix, $\mathbb{R}^{\lvert L_b\rvert}$	Broker seed fitting batch using the selected input map.

Local variables for InitializePrincipal

Symbol	Dimension/domain	Definition
σ_x	scalar	Principal type noise scale.
u_i	scalar	Principal i curve position.
ϵ_i	$\mathbb{R}^d$	Principal type noise for principal i.
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N]\setminus\{i\}$	Entrant attachment set.

Algorithm 5 InitializePrincipal

Require: State S, principal slot i, period counter t, parameters ϑ.
Ensure: Principal slot i is initialized in S.

Draw $u_i \sim \text{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
$x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{i\cdot}^t$, $C_{i\cdot}^t$, χ_i^t, M_i^t, and Δ_i^t.
Initialize the selected predictor to the constant Q: for NN, draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$, $\mathcal{O}_i^t = 0$; for ridge, set $\alpha_i^t = Q$ and $\beta_i^t = 0$.
if $t = 0$ **then**
    $s_i^t \leftarrow (0,0)$.
else
    Remove i from Roster^t, D^t, and M_j^t for all j; delete all incident edges in G^t; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
    Sample $\mathcal{V}_i^{\text{ent}} \subset [N]\setminus\{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

    Add edges (i,j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.
    $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j\in\mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$
    $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
    Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 6 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \mathrm{Unif}\{1,\dots,5\}$ and phases $\psi_\ell \sim \mathrm{Unif}[0,2\pi)$ for $\ell = 1,\dots,d_\gamma$, defining γ.

Allocate empty principal and broker containers in S^0 and store γ.

for $i = 1,\dots,N$ **do**

    InitializePrincipal$(S^0, i, 0, \vartheta)$.

Draw $u_c \sim \mathrm{Unif}[0,1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow [\gamma(u_c) + \epsilon_c]/\|\gamma(u_c) + \epsilon_c\|_2$.

Draw $A_0 \in \mathbb{R}^{d\times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d/\operatorname{tr}(A_0^\top A_0)$.

$\Sigma_x \leftarrow N^{-1}\sum_i x_i^0 (x_i^0)^\top$.

Draw symmetric zero-trace B_0 and set

$$B_{\mathrm{raw}} \leftarrow B_0 - \frac{\operatorname{tr}(\Sigma_x B_0 \Sigma_x A)}{\operatorname{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\mathrm{raw}}/\|B_{\mathrm{raw}}\|_F.$$

Calibrate $(\bar{q}_{\mathrm{cal}}, r, \phi, c_s)$.

Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.

Initialize $\mathcal{H}_b^0$ and the selected broker predictor to the constant Q using the corresponding NN or ridge initialization above; set $\mathrm{rep}_b^0 = 0$.

Sample $\mathrm{Roster}^0 \subset [N]$ uniformly with $|\mathrm{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.

for each unordered non-broker edge $\{i,j\} \in \mathcal{E}_G^0$ **do**

    Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.

Let $\mathcal{B}_{\mathrm{seed}} \leftarrow \{\{i,j\} \in \mathcal{E}_G^0 : i,j \in \mathrm{Roster}^0\}$; shuffle and keep at most 100 edges.

for each $\{i,j\} \in \mathcal{B}_{\mathrm{seed}}$ **do**

    Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0.

    If $\mathsf{L} = \mathrm{Ridge}$ and $a_b = \text{single-principal}$, draw and store $r_\ell \sim \mathrm{Unif}\{1,2\}$ for this observation.

if $n_b^0 > 0$ **then**

    Set rep_b^0 to mean broker seed output.

else

    Set $\mathrm{rep}_b^0 \leftarrow 0$.

Set $s_{i,\mathrm{self}}^0$ to mean own seed output and $s_{i,\mathrm{broker}}^0 \leftarrow \mathrm{rep}_b^0$.

Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every principal and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.

for each i with $n_i^0 > 0$ **do**

    $L_i \leftarrow \{1,\dots,n_i^0\}$; apply the observation cap as in Algorithm 4.

    $\Theta_i^0 \leftarrow \mathrm{FitPredictor}(\mathsf{L}, a, \Theta_i^0, (\Xi_i^0)_{\cdot L_i}, (y_i^0)_{L_i}, \lambda_a, n_i^0, n_i^0, \vartheta)$; $\Delta_i^0 \leftarrow 0$.

if $n_b^0 > 0$ **then**

    $L_b \leftarrow \{1,\dots,n_b^0\}$; apply the same observation cap and index selection.

    If $\mathsf{L} = \mathrm{Ridge}$ and $a_b = \text{size-matched}$, apply the size-matched rule above to L_b.

    $Z_{b,\mathrm{seed}} \leftarrow [z_{b\ell}]_{\ell \in L_b}$, $y_{b,\mathrm{seed}} \leftarrow (y_b^0)_{L_b}$.

    $\Theta_b^0 \leftarrow \mathrm{FitPredictor}(\mathsf{L}, b, \Theta_b^0, Z_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$; $\Delta_b^0 \leftarrow 0$.

return S^0.

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Principal i state (s_i, χ_i), broker reputation rep_b, RNG.

$U_{\text{self}} \leftarrow s_{i,\text{self}}$.
$U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1-\chi_i)\text{rep}_b$.
if $U_{\text{self}} > U_{\text{broker}}$ **then**
    return self.
if $U_{\text{broker}} > U_{\text{self}}$ **then**
    return broker.
return self w.p. $1/2$, else broker ▷ $U_{\text{self}} = U_{\text{broker}}$.

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η	scalar	Principal exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.
Ensure: Updated state S^t, accepted records $\mathcal{R}^t$, and pre-turnover measure row Y^t.

Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.
Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil \alpha_R N \rceil$.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
$\mathcal{D}^t \leftarrow \emptyset$.
for $i = 1, \ldots, N$ **do**
    Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
    if $d_i^t > 0$ **then**
        $z_i^t \leftarrow \textsc{ChooseChannel}(i, \text{rep}_b^{t-1})$.
        Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
        if $z_i^t =$ broker **then**
            Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.

TrainPredictors(S, t, ϑ).
$\mathcal{R}^t \leftarrow$ OfferMarket$(S, \mathcal{D}^t, \vartheta)$.
Synchronize broker-node edges to Roster^t, D^t, and current broker-channel matches.
UpdateSatisfaction$(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
    $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s^t_{i,\text{broker}}$.
else
    $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
if $t \bmod \Delta_{\text{net}} = 0$ **then**
    Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t.
$Y^t \leftarrow$ CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
Append n_i^t to $\mathbf{p}_i^t$ for every principal and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
for $i = 1, \ldots, N$ **do**
    if principal i exits with probability η **then**
        InitializePrincipal(S^t, i, t, ϑ).
Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
return $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Principal predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_B} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i.
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i\to j}^{\text{self},t}$	scalar	Self-offer score.
$\mathcal{U}_{b,i}$	subset of $[N]$	Candidate set for broker-client demander i.
$v_{b,i\to j}^t$	scalar	Score used to order brokered candidates for demander i.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{I}_{b,i}$	list	Brokered candidates for client i, sorted by $v_{b,i\to j}^t$.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i\to j}^{\text{self}}, u_{b,i\to j}^{\text{local}}, u_{b,i\to j}^{\text{global}}$	$[0, 1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and o.from.

Local variables for OfferMarket (continued)

Symbol	Dimension/domain	Definition
$z^\star$	channel	Accepted-record channel.
ι	$\{0,1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i,j) = \textsc{PredictBroker}(i,j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i,j) \in \mathcal{E}_G^t\}$ exclude the broker node. Principal i assesses any presented candidate as

$$a_{i\to j}^t = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \textsc{PredictPrincipal}(i, x_j^t), & \text{otherwise.} \end{cases}$$

Self-search makes this score available only for neighbors and non-neighbors in the demander-specific stranger pool:

$$v_{i\to j}^{\text{self},t} = \begin{cases} a_{i\to j}^t, & j \in \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i)), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i,j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \textsc{PredictPrincipal}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i,j,z,v).
if O_{ij}^t is defined **then**
  return 0.
$O_{ij}^t \leftarrow (i,j,z,v)$.
if O_{ji}^t is undefined **then**
  $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i,j\}\}$.
return 1.

Algorithm 10 BrokerCandidates

Require: Broker-client demander i, state S, broker service S_b.
Ensure: Candidate set $\mathcal{U}_{b,i}$ and score function $v_{b,i\to j}^t$.
if $\mathsf{S}_b =$ full **then**
  $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else if $\mathsf{S}_b =$ assessment-only **then**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
  $\mathcal{U}_{b,i} \leftarrow [\mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))] \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else

$\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow a_{i\to j}^t$. ▷ access-only

return $(\mathcal{U}_{b,i}, v_{b,i\to j}^t)$.

Algorithm 11 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.

For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
$O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
for each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
    Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
    $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
    Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
    Draw independent $u_{i\to j}^{\text{self}} \sim \text{Unif}(0,1)$ for each feasible j.
    $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}:\ v_{i\to j}^{\text{self},t} > r}(-v_{i\to j}^{\text{self},t}, u_{i\to j}^{\text{self}})$.
    for the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ principals j in $\mathcal{I}_i^{\text{self}}$ **do**
        AddOffer$(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i\to j}^{\text{self},t})$.
$\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
$D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t,\ \Omega_i > 0\}$.
$\mathcal{B}_b^t \leftarrow \emptyset$.
for each $i \in D_+^t$ **do**
    $(\mathcal{U}_{b,i}, v_{b,i\to j}^t) \leftarrow$ BrokerCandidates(i, S, S_b).
    Draw independent $u_{b,i\to j}^{\text{local}} \sim \text{Unif}(0,1)$ for each feasible j.
    $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{U}_{b,i}:\ v_{b,i\to j}^t > r}(-v_{b,i\to j}^t, u_{b,i\to j}^{\text{local}})$.
    for the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ principals j in $\mathcal{I}_{b,i}$ **do**
        Append $(i, j, v_{b,i\to j}^t)$ to $\mathcal{B}_b^t$.
Draw a new independent $u_{b,i\to j}^{\text{global}} \sim \text{Unif}(0,1)$ for each $(i, j, v) \in \mathcal{B}_b^t$.
Sort $\mathcal{B}_b^t$ lexicographically by $(-v, u_{b,i\to j}^{\text{global}})$.
for each $(i, j, v) \in \mathcal{B}_b^t$ **do**
    if $\Omega_i > 0$ **then**
        $\iota \leftarrow$ AddOffer$(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$.
        $\Omega_i \leftarrow \Omega_i - \iota$.
for each $\{i, j\} \in \mathcal{P}_O^t$ **do**
    $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
    if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
    else if $o_{ij} \neq \emptyset$ and $w_{j\leftarrow i}^t > r$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
    else if $o_{ji} \neq \emptyset$ and $w_{i\leftarrow j}^t > r$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 12 FinalizeMatch

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.

if $j \in M_i^t$ **then**

return .
Draw $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
Append columns x_j^t to Ξ_i and x_i^t to Ξ_j; append q_{ij}^t to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i, j) to G^t; add j to M_i^t and i to M_j^t.
if $z =$ broker or ($o' \neq \emptyset$ and $o'.z =$ broker) **then**
    Append the two party types x_i^t and x_j^t to $X_{b,1}$ and $X_{b,2}$; append q_{ij}^t to y_b; increment n_b.
    If $\mathsf{L} =$ Ridge and $a_b =$ single-principal, draw and store $r_\ell \sim \text{Unif}\{1, 2\}$ for this observation.
    $z^\star \leftarrow$ broker.
else
    $z^\star \leftarrow$ self.
Append accepted record $(i, j, z^\star, q_{ij}^t, v, o, o')$ to $\mathcal{R}^t$.

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Principal i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i.
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 13 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.
for each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i\}\, \zeta.q$.
    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i,\ o.z =$ broker$\}$.
    if $z_i =$ self **then**
        $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s$.
    else
        $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i$; $\chi_i^t \leftarrow 1$.
    $s_{i,z_i}^t \leftarrow (1 - \omega) s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t$.

CollectMeasures. CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ records the measures defined in Appendix B, Section 8, without changing the simulation state.

Appendix B
Model Specifications for an ABM of Brokerage in Matching Markets

This document specifies an agent-based model of brokered matching.

The model is a discrete-time simulation of a matching market with *principals* and a *broker*. Principals seeking pairwise matches either search on their own or outsource to the broker. All economic quantities (match output, fees, surplus) are in the same payoff units. All principals use heuristic decision rules. No principal holds beliefs about other principals' strategies.

The model specification covers principals (§1), the matching problem (§2), how principals learn to predict match value (§3), match economics (§4), network structure and principal turnover (§5), search and broker access (§6), the outsourcing decision (§7), and period measures (§8). Baseline parameter values and reproducibility are documented in Part II (§9).

Part I. Model Specification

1. Principals and Time

The model runs for T discrete periods (default 500). It has N principals (default 1000), a single broker, and an undirected network G over the principals and broker.

Each principal i is characterized by:

- **Type** $\mathbf{x}_i \in \mathbb{R}^d$: a fixed vector of observable characteristics, drawn as specified below and used by the matching function (§2).
- **Current-period matches** M_i^t: active bilateral relationships involving i in period t. Each counterparty can appear at most once in M_i^t; incoming accepted offers are not capped.
- **Active demand bound** K: the maximum number of active demands a principal can initiate in a period (default $K = 5$; §6).
- **Experience history** $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}$: (other party's type, realized match output) pairs from all matches involving i (§3a).
- **Satisfaction indices** $s_{i,c}^t$: one scalar per search channel $c \in \{\text{self}, \text{broker}\}$, used in the outsourcing decision (§7).

Principal Types

Type vectors are multidimensional but concentrated near a smooth one-dimensional curve on the unit sphere in $\mathbb{R}^d$. Let $u \in [0, 1]$ index position along the curve:

$$\boldsymbol{\gamma}(u) = \frac{\tilde{\boldsymbol{\gamma}}(u)}{\|\tilde{\boldsymbol{\gamma}}(u)\|}, \qquad \tilde{\gamma}_\ell(u) = \begin{cases} \sin(2\pi\nu_\ell u + \psi_\ell) & \ell = 1, \ldots, d_\gamma \\ 0 & \ell = d_\gamma + 1, \ldots, d. \end{cases}$$

The frequencies $\nu_\ell \sim \text{Unif}\{1, 2, 3, 4, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ are drawn once per simulation. The parameter $d_\gamma \leq d$ is the number of coordinates in which the curve varies.

Each principal is drawn at a random position $u_i \sim \text{Unif}[0, 1]$ on the curve, then perturbed:

$$\mathbf{x}_i = \frac{\boldsymbol{\gamma}(u_i) + \boldsymbol{\epsilon}_i}{\|\boldsymbol{\gamma}(u_i) + \boldsymbol{\epsilon}_i\|}, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right).$$

Noise is applied in all d dimensions, then the vector is re-normalized. The scale $\sigma_x/\sqrt{d}$ keeps the expected displacement from the curve approximately σ_x for any d.

Thus the curve remains one-dimensional for every d_γ. Increasing d_γ changes how many coordinates contain systematic curve variation; the remaining coordinates contain only perturbation noise.

Broker

The broker is a permanent node in G and has no type vector. Its state is:

- **Experience history** $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}$: (party 1 type, party 2 type, realized match output) triples from mediated matches (§3c). The party labels have no directional meaning because the broker's pair features are symmetric.
- **Roster** Roster^t: the broker's standing access base (§6b).
- **Current clients** D^t: principals who outsource to the broker in period t (§6b).
- **Reputation** rep_b^t: the average broker satisfaction of current clients (§7).

2. The Matching Problem

Each potential pair's underlying match value combines two components: the average of the principals' individual-quality scores and complementarity between their types. Each principal's individual-quality score is the same across partners, whereas complementarity depends on the pairing. The parameter ρ controls the relative contributions of these components. The parameter δ makes complementarity harder to predict by increasing the difference between otherwise similar pairs on opposite sides of a boundary in type-pair space. A pair's underlying match value does not depend on which principal initiates the match. Both principals observe the same realized match output.

2a. Match Output

Let q_{ij} be the realized per-period output of a match between principals i and j, measured in payoff units:

$$q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

A pair's underlying match value is $Q + f(\mathbf{x}_i, \mathbf{x}_j)$, determined by the two types and the matching-function objects fixed at initialization. Because the shock ε_{ij} is independent and has mean zero, underlying match value also equals $E[q_{ij} \mid \mathbf{x}_i, \mathbf{x}_j]$, holding the matching environment fixed. The shock creates variation in realized match output that neither the principals nor the broker can predict.

The function $f : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is unknown to all principals:

$$f(\mathbf{x}_i, \mathbf{x}_j) = \frac{1}{s_{\text{mix}}(\rho, \delta)} \left[\sqrt{\rho}\, \widetilde{U}_{ij} + \sqrt{1-\rho}\, \widetilde{W}_{ij}(\delta) \right].$$

The tildes indicate that the individual-quality component U_{ij} (§2b) and the complementarity component $W_{ij}(\delta)$ (§2c) are centered and scaled using fixed moments. These fixed moments are calculated across all unordered pairs of distinct principals within each of 50 independently generated calibration markets of 1,000 principals, then averaged with equal weight across markets. The resulting coefficients depend on ρ and δ, are shared across simulations, and remain fixed through turnover.

Let $\kappa(\delta)$ denote the average within-market covariance between the standardized individual-quality and complementarity components in the calibration markets. The common adjustment

$$s^2_{\text{mix}}(\rho, \delta) = 1 + 2\sqrt{\rho(1-\rho)}\, \kappa(\delta)$$

accounts for this covariance. Each standardized component and the combined systematic component f have unit average within-market variance in the calibration markets for every ρ and δ. Realized within-market variance of f can differ from one and change with turnover. The calibration normalization defines the payoff unit. In these units, $Q = 2.84$ and $\sigma_\varepsilon = 0.28$.

The standardized components have equal average within-market variances across the calibration markets. On this calibration scale, after weighting, the individual-quality component accounts for a share ρ of their summed marginal variances, and complementarity accounts for $1 - \rho$. At $\rho = 0$, match value depends only on complementarity; at $\rho = 1$, it depends only on individual quality. At $\rho = 1$, δ has no effect.

2b. Individual Quality

The vector $\mathbf{c}$ defines a common ideal type and is fixed within each simulation. Principal i's individual-quality score, $\mathbf{x}_i^\top \mathbf{c}$, measures its alignment with that ideal. A pair's individual-quality component is the average of the two principals' scores:

$$U_{ij} = \frac{1}{2}(\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c}), \qquad \widetilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}.$$

The ideal type $\mathbf{c} \in \mathbb{R}^d$ is drawn at initialization as a perturbation of a random point on the principal type curve:

$$u_c \sim \text{Unif}[0,1], \qquad \boldsymbol{\epsilon}_c \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d}\mathbf{I}_d\right), \qquad \mathbf{c} = \frac{\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c}{\|\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c\|_2}.$$

The normalization places $\mathbf{c}$ on the same unit sphere as principal types. The perturbation therefore changes the ideal direction without changing its magnitude.

2c. Match-Specific Complementarity

Define base complementarity, signed complementarity, and difficulty-adjusted complementarity as

$$V_{ij} = \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j, \qquad H_{ij} = \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j) V_{ij}, \qquad W_{ij}(\delta) = V_{ij} + \delta H_{ij}.$$

The normalized complementarity used in the matching function is

$$\widetilde{W}_{ij}(\delta) = \frac{W_{ij}(\delta) - \mu_W(\delta)}{\sigma_W(\delta)}.$$

A is a symmetric positive definite matrix that gives a smooth bilinear complementarity geometry. Because V_{ij} depends jointly on both types, it cannot in general be represented as the sum of separate principal scores. A second symmetric matrix, **B**, assigns pairs to two regions of type-pair space. Both matrices are fixed for the duration of the simulation.

Base complementarity. Draw $\mathbf{M}_A \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0,1)$ entries and set

$$\mathbf{A} = \mathbf{M}_A^\top \mathbf{M}_A \cdot \frac{d}{\text{tr}(\mathbf{M}_A^\top \mathbf{M}_A)}.$$

Thus **A** is symmetric positive definite and $\text{tr}(\mathbf{A}) = d$.

The trace normalization prevents arbitrary scale differences in **A** across realizations. The fixed pair-level normalization described in §2a then places complementarity on the common scale used in the matching function.

Difficulty parameter. Equivalently, write $W_{ij}(\delta) = g(\mathbf{x}_i, \mathbf{x}_j) V_{ij}$, where a second symmetric matrix $\mathbf{B} \in \mathbb{R}^{d \times d}$ defines

$$g(\mathbf{x}_i, \mathbf{x}_j) = 1 + \delta \cdot \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)$$

where $\delta \in [0, 1]$ (default 0.5).

The multiplier is $1 + \delta$ when $\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j > 0$, $1 - \delta$ when it is negative, and 1 when it is zero. At $\delta = 0$, the boundary between the two regions has no effect. For $\delta > 0$, pairs on opposite sides of the boundary can have sharply different underlying match values even when their types are otherwise similar. Increasing δ increases this difference, making complementarity harder to learn. The scale $\sigma_W(\delta)$ holds the average within-market variance of standardized complementarity at one in the calibration markets as δ varies.

Boundary construction. At initialization, **B** is made orthogonal to **A** under the weighted inner product defined by the initial principals' empirical second-moment matrix. The resulting boundary remains fixed through turnover. Let

$$\mathbf{\Sigma}_x = \frac{1}{N} \sum_{i=1}^{N} \mathbf{x}_i \mathbf{x}_i^\top$$

be the uncentered second-moment matrix of the initial type vectors.

Draw $\mathbf{Z} \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0,1)$ entries and form a symmetric zero-trace matrix:

$$\widetilde{\mathbf{B}} = \frac{\mathbf{Z} + \mathbf{Z}^\top}{2}, \qquad \mathbf{B}_0 = \widetilde{\mathbf{B}} - \frac{\operatorname{tr}(\widetilde{\mathbf{B}})}{d}\mathbf{I}_d.$$

Define the weighted inner product

$$\langle \mathbf{M}, \mathbf{N} \rangle_{\boldsymbol{\Sigma}_x} = \operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{M} \boldsymbol{\Sigma}_x \mathbf{N}),$$

then remove the projection of $\mathbf{B}_0$ onto $\mathbf{A}$ and normalize:

$$\mathbf{B}_{\text{raw}} = \mathbf{B}_0 - \frac{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B}_0 \boldsymbol{\Sigma}_x \mathbf{A})}{\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A})}\mathbf{A}, \qquad \mathbf{B} = \frac{\mathbf{B}_{\text{raw}}}{\|\mathbf{B}_{\text{raw}}\|_F}.$$

This gives $\operatorname{tr}(\boldsymbol{\Sigma}_x \mathbf{B} \boldsymbol{\Sigma}_x \mathbf{A}) = 0$ and $\|\mathbf{B}\|_F = 1$. Weighting by $\boldsymbol{\Sigma}_x$ removes overlap between the two bilinear scores under the initial empirical type distribution, rather than in unused regions of $\mathbb{R}^d$.

Symmetry of $\mathbf{A}$ and $\mathbf{B}$ makes $f(\mathbf{x}_i, \mathbf{x}_j) = f(\mathbf{x}_j, \mathbf{x}_i)$.

3. Learning

Before searching for matches, principals and the broker may update their prediction models on their accumulated histories. Updated models are used to score candidates (§6). The broker is eligible to refit every period. Principal i is eligible when $i \bmod 2 = t \bmod 2$, so each principal is eligible every other period. An eligible model refits only when its history is nonempty and it has received new observations since its previous fitting call. A parameter switch selects the prediction model. Neural network (NN) learning is the default. Ridge regression is the alternate experiment.

3a. Prediction Models and Fitting

In the default model, both principals and the broker use a fully connected network with one hidden layer, ReLU activations, and a single linear output:

$$\hat{q}(\mathbf{z}) = \mathbf{w}_2^\top \operatorname{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + b_2$$

where $\hat{q}$ is predicted match value, $\mathbf{z}$ is the input feature vector and the remaining symbols are network weights and biases. Principals receive raw partner type vectors; the broker receives pairs of them.

Training rule. Network weights minimize mean squared error (MSE) with the Adam optimizer on the current training window. Principals and the broker may have separate learning rates, $\eta_{\text{lr},a}$ and $\eta_{\text{lr},b}$. No explicit weight regularization is applied. At $t = 0$, each network sets $\mathbf{W}_1$ using He initialization, $\mathbf{b}_1 = \mathbf{0}$, $\mathbf{w}_2 = \mathbf{0}$, and $b_2 = Q$ (§2a). An untrained predictor therefore returns exactly Q for every input. Networks with seed observations then take $E_{\text{init},c}$ gradient steps on their seed histories, where $c \in \{a, b\}$.

Alternate ridge rule. In the ridge experiment, principals and the broker instead use

$$\hat{q}_i(\mathbf{x}_j) = \alpha_i + \boldsymbol{\beta}_i^\top \mathbf{x}_j, \qquad \hat{q}_b(\mathbf{x}_i, \mathbf{x}_j) = \alpha_b + \boldsymbol{\gamma}_b^\top \psi(\mathbf{x}_i, \mathbf{x}_j).$$

Principal and broker fits minimize mean squared error plus λ_a and λ_b, respectively, times the squared Euclidean norm of the slope coefficients. Intercepts are not penalized. Raw type and pair

features are used without sample standardization. Before its first fit, a ridge predictor has zero slopes and intercept Q. The reporting simulations use $\lambda_a = 0.001$ and $\lambda_b = 0.001$.

Subsequent NN updates warm-start from the previous weights and persistent Adam moment state. The number of gradient steps adapts to the ratio of new observations to total history:

$$E_{c,t} = \max\left(E_{\text{steps},c},\ \left\lceil E_{\text{init},c} \cdot \frac{n_{\text{new}}}{n_{\text{total}}} \right\rceil\right), \qquad c \in \{a, b\}.$$

where n_{new} is the number of observations added since that model's previous fitting call, $n_{\text{total}} = |\mathcal{H}^t|$ is full history size, and $E_{\text{steps},c}$ is the learner-specific minimum per-call step count. For a principal, n_{new} can span multiple periods because principals fit on an alternating schedule. Principals and the broker use the same functional rule but may have different parameter values. Ridge is solved directly at each fitting call. The reporting NN values are $\eta_{\text{lr},a} = 0.003$, $\eta_{\text{lr},b} = 0.03$, $E_{\text{init},a} = E_{\text{init},b} = 100$, and $E_{\text{steps},a} = E_{\text{steps},b} = 50$.

Initialization is period 0 for each learner. Its seed-history count is stored as the first period boundary. Each later boundary stores cumulative history at the end of a simulation period. A fitting call uses observations from the most recent $W_p = 40$ completed simulation periods. Before 40 simulation periods are available, the window also includes period 0. Once 40 are available, the period-0 seed history is excluded. If the selected history block exceeds $M_{\text{train}} = 2000$ observations, fitting deterministically keeps 2000 spaced observations across that block, including its oldest and newest observations. If the cap is one, it keeps the newest observation. The cap changes only the fitting sample, not the stored history or the period window. NN and ridge use the same window, cap, fitting cadence, and histories.

3b. Principal i's Model

History. $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}_{m=1}^{n_i}$ records the other party's type and realized match output from every match involving i.

Input. The principal's network takes the partner's type as input: $\mathbf{z} = \mathbf{x}_j$.

The hidden width is $h_A = 2d$, so the baseline uses 16 hidden units.

3c. Broker's Model

History. $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}_{m=1}^{n_b}$ records both parties' types and realized match output from broker-mediated matches.

Input. The broker's network takes a symmetric pair feature vector:

$$\psi(\mathbf{x}_i, \mathbf{x}_j) = \left[\mathbf{x}_i + \mathbf{x}_j;\ \text{vech}\left(\frac{\mathbf{x}_i\mathbf{x}_j^\top + \mathbf{x}_j\mathbf{x}_i^\top}{2}\right)\right].$$

The half-vectorization vech keeps the lower-triangular entries of the symmetric matrix, so the broker input dimension is $d_B = d + d(d+1)/2$, equal to 44 at $d = 8$.

The broker hidden width is $h_B = 8d$, so the baseline uses 64 hidden units.

Because $\psi(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_j, \mathbf{x}_i)$, each observed unordered pair contributes one training row.

The broker learns only from mediated matches. It does not observe self-search outcomes.

3d. Ridge Broker Ablations

Three separate ridge-only ablations change the broker while leaving every principal's local ridge model unchanged. **Size-matched pooled history** uniformly samples the broker's windowed and capped fitting data without replacement, using the smaller of its available sample size and the nearest-integer median sample size among principals with nonempty histories, with half-integers rounded up. **Single-principal history** randomly retains one endpoint when each broker observation is recorded, fits $g_b(\mathbf{x}) = \alpha_b + \boldsymbol{\gamma}_b^\top \mathbf{x}$, and scores a pair as $g_b(\mathbf{x}_i) + g_b(\mathbf{x}_j) - \bar{q}_b$, where $\bar{q}_b$ is the current fitting-sample output mean and defaults to Q before fitting. **Additive ridge** uses $\mathbf{x}_i + \mathbf{x}_j$ without bilinear features. All use the calibrated principal and broker penalties, period window, observation cap, and fitting cadence.

4. Match Economics

Matches use a common reservation threshold and channel-specific cost accounting.

At initialization, each simulation computes $\bar{q}_{\text{cal}}$ as the sample mean of $Q + f(\mathbf{x}_i, \mathbf{x}_j)$ over 10,000 pairs whose indices are drawn independently and uniformly with replacement from the initial principals. This quantity scales the reservation threshold, broker fee, and self-search cost.

4a. Outside Options

All principals share a reservation threshold r, calibrated at initialization:

$$r = f_r \cdot \bar{q}_{\text{cal}}$$

where $\bar{q}_{\text{cal}}$ is the calibration mean and f_r is the reservation coefficient (default 0.60). Since ϕ and c_s do not depend on r (§4c), f_r may take any nonnegative value, including $f_r \geq 1$, and is swept as an analysis axis.

4b. Participation Constraints

Demanders emit offers only when the candidate evaluation exceeds r. In self-search, this evaluation is the pair's mean past realized output $\bar{q}_{ij}$ for neighbors with prior pair-specific experience and the demander prediction $\hat{q}_i(\mathbf{x}_j)$ otherwise (§6a). Brokered search uses the broker pair prediction $\hat{q}_b(\{i, j\})$. A one-sided offer is accepted only if the receiver's evaluation exceeds r; reciprocal offers are accepted automatically (§6d).

4c. Search Costs

The self-search cost and broker fee share one calibrated level:

$$\phi = c_s = \lambda_c \cdot \bar{q}_{\text{cal}}.$$

λ_c is the shared cost rate (default 0.05). The two symbols differ in payoff accounting:

- **Self-search** costs c_s on each requested self-search position, whether or not it is filled.
- **Brokerage** charges ϕ only on successful brokered placements.

5. Network Structure and Turnover

Principals interact through a single undirected network G that determines search opportunities and structural position.

5a. Network Initialization

G is initialized as a Watts-Strogatz small-world graph on the N principals (Watts & Strogatz, 1998), with ring degree $k_G = 6$, rewiring probability $p_{\text{rewire}} = 0.1$, and random principal order on the ring.

The broker is added as a permanent node with no type vector. Broker-node edges are synchronized to standing roster members, current-period broker clients, and principals currently engaged in broker-channel matches (§6b).

5b. Match Tie Formation

Each realized match adds an undirected i-j edge if one does not already exist. Ties persist until either node exits.

5c. Principal Turnover

Principals exit independently each period with probability η (default 0.02). An exiting principal is removed everywhere. Its node slot is reused by an entrant with a fresh type drawn from the same type process (§1). The entrant receives $\lfloor k_G/2 \rfloor$ edges to existing principals sampled with probability proportional to $\exp(-\|\mathbf{x}_{i'} - \mathbf{x}_j\|^2)$.

6. Search

At the start of each period, each principal draws active relationship demand

$$d_i \sim \text{Binomial}(K,\ p_{\text{demand}}),$$

where K is the maximum number of active demands the principal can initiate in the period and p_{demand} defaults to 0.50. If $d_i > 0$, the principal chooses **one channel for the batch** (§7): self-search or broker.

6a. Self-Search

Principal i's self-search candidate pool has two components:

Neighbors. For a direct network neighbor j with prior pair-specific experience, principal i uses $\bar{q}_{ij} = \frac{1}{n_{ij}} \sum q_{ij}^{(m)}$, where n_{ij} is the number of times i and j have matched. If the graph edge exists but $n_{ij} = 0$, the principal instead uses $\hat{q}_i(\mathbf{x}_j)$. Initial histories seed one observation for every initial graph edge. Unobserved neighbor edges arise after turnover because entrants attach structurally before matching.

Strangers. Each self-searching demander i independently samples a pool U_i^t of $\min(n_{\text{strangers}}, N)$ principals uniformly without replacement, where $n_{\text{strangers}} = 10$ (default). The demander evaluates pool members that are not current neighbors using $\hat{q}_i(\mathbf{x}_j)$ (§3b).

Principal i orders feasible self-search candidates by these evaluations and emits up to d_i directed offers to candidates above r. Equal evaluations are ordered uniformly at random, so a principal whose predictions are constant produces a random ranking rather than an undefined ordering.

6b. Broker Roster and Access

The broker maintains a **roster** of principals it can propose as counterparties when mediating matches. In each period, it also observes the current-client set D^t of principals who outsourced. Broker access for the period is:

$$\mathcal{A}_b^t = \text{Roster}^t \cup D^t.$$

Current clients expand broker access for the period but do not become standing roster members.

Initialization. The roster is seeded with a target size

$$R^* = \lceil \alpha_R N \rceil,$$

where the broker roster share α_R defaults to 0.20. The model draws R^* principals uniformly at random (default 200 at $N = 1000$). The broker's history is seeded from existing roster-roster ties in the initial graph, capped at 100 observed ties.

Roster refresh. At the start of each period, each member independently leaves the roster with probability p_{roster} (default 0.02). If $\widetilde{\text{Roster}}^t$ is the post-churn roster,

$$\widetilde{\text{Roster}}^t = \{i \in \text{Roster}^{t-1} : u_i^t > p_{\text{roster}}\}, \qquad u_i^t \overset{iid}{\sim} \text{Unif}[0,1],$$

then the broker samples without replacement from $\{1, \ldots, N\} \setminus \widetilde{\text{Roster}}^t$ until $|\text{Roster}^t| = R^*$, or until the population is exhausted. Broker-node edges are synchronized to the standing roster, current clients, and current broker-mediated matches (§5a).

6c. Broker-Mediated Search

For every unordered pair $\{i, j\}$ such that one side is a broker-client demander and the other side is in $\mathcal{A}_b^t$, the broker computes predicted match value. For each broker-client demander i, it ranks accessible counterparties above r, breaking equal predictions uniformly at random, and keeps the highest-ranked candidates up to i's residual active broker demand. Directed broker offers are then processed in a global score ranking with a new, independent random ordering within score ties. If both sides are broker-client demanders with remaining active demand, the same unordered pair can generate reciprocal broker offers.

Assessment-only. The broker scores the principal's network neighbors and an independently sampled stranger pool (§6a) using its pair predictions.

Access-only. Candidates come from the broker access set $\mathcal{A}_b^t$. Scores use the requesting principal's pair-specific mean for experienced neighbors and its prediction model otherwise (§6a).

All other rules remain as specified. The broker still records mediated matches and updates its prediction model (§3), but its predictions do not determine access-only offers.

6d. Offer Acceptance and Match Formation

All self-search and broker offers enter one shared market, grouped by unordered pair.

1. If pair $\{i, j\}$ contains reciprocal offers $i \to j$ and $j \to i$, the relationship is accepted automatically.
2. If pair $\{i, j\}$ contains one directed offer $i \to j$, receiver j evaluates i using $\bar{q}_{ji}$ for known partners and $\hat{q}_j(\mathbf{x}_i)$ otherwise. The offer is accepted iff this value exceeds r.
3. Each accepted unordered pair forms at most one current-period relationship.

At market finalization, for each accepted match: 1. Realized output is drawn: $q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}$. 2. Both parties add the observation to their histories: i adds $(\mathbf{x}_j, q_{ij})$ to $\mathcal{H}_i$; j adds $(\mathbf{x}_i, q_{ij})$ to $\mathcal{H}_j$. 3. If either directed offer used the broker channel, the broker adds $(\mathbf{x}_i, \mathbf{x}_j, q_{ij})$ to $\mathcal{H}_b$ once. 4. An edge is added between i and j in G (if not already present).

A demander can have at most d_i outgoing offers, but can receive any number of acceptable incoming offers. Thus K is an active-demand bound, not a capacity constraint on receiving offers.

Before the next period begins, match lists are cleared. The next period begins with no current-period matches.

7. The Outsourcing Decision

7a. Satisfaction Tracking

Each principal i maintains a satisfaction index $s^t_{i,c}$ for each search channel $c \in \{\text{self}, \text{broker}\}$. For the chosen channel, the index updates once per active-demand period as an exponentially weighted moving average (EWMA) of realized match output net of search costs:

$$s^{t+1}_{i,c} = (1 - \omega)\, s^t_{i,c} + \omega \cdot \tilde{q}$$

where $\omega = 0.2$ and $\tilde{q}$ is the period input in the table. The denominator is requested demand d_i, so unfilled requests contribute zero output. A reciprocal relationship can update two channel-specific states, one for each directed offer.

Channel	Satisfaction input $\tilde{q}$
Self-search	$\dfrac{\sum q_{ij}}{d_i} - c_s$, summing over accepted directed offers made by i through self-search
Brokered	$\dfrac{\sum q_{ij} - \phi \cdot n_{i,\text{broker success}}}{d_i}$, summing over accepted directed offers made by i through the broker

Brokered failure gives $\tilde{q} = 0$; self-search failure gives $\tilde{q} = -c_s$.

$s^t_{i,\text{self}}$ covers all self-search, including known neighbors.

At initialization, satisfaction with self-search is the mean of seed match outcomes and satisfaction with brokerage is the broker's seed-data reputation (§7c). Fresh entrants set satisfaction with self-search to the neighbors' mean for that channel and satisfaction with brokerage to current broker reputation.

The first broker choice, regardless of placement success, switches the principal from broker reputation to its own $s^t_{i,\text{broker}}$ (§7b).

7b. Decision Rule

Each principal with $d_i > 0$ compares:

- $\textbf{score}_{\textbf{self}} = s^t_{i,\text{self}}$
- $\textbf{score}_{\textbf{broker}} = s^t_{i,\text{broker}}$ if the principal has tried the broker, otherwise rep^t_b.

The principal chooses the higher-score channel. Ties are resolved by a fair coin flip.

7c. Broker Reputation

$$\text{rep}^{t+1}_b = \begin{cases} \frac{1}{|D^t|} \sum_{i \in D^t} s^{t+1}_{i,\text{broker}} & \text{if } D^t \neq \emptyset \\ \text{rep}^t_b & \text{otherwise} \end{cases}$$

where D^t is the set of principals who outsourced to the broker this period. With current clients, reputation is their mean post-update broker satisfaction; with no clients, it is unchanged. Reputation is initialized from the mean broker seed outcome.

8. Period Measures

Period measures are evaluated after current-period matching, satisfaction, and reputation updates, but before principal entry/exit turnover. Principal decisions do not depend on these measures.

Network Measures

Broker network-position measures use G, including the broker node (§5a). They are computed when $t \bmod \Delta_{\text{net}} = 0$ (default 20). Other period measures are recorded each period.

Betweenness centrality. Freeman betweenness (Freeman, 1977) for the broker node:

$$\text{BC}_b = \frac{1}{(n-1)(n-2)} \sum_{u \neq b} \sum_{v \neq u} \frac{\sigma_{uv}(b)}{\sigma_{uv}}$$

where $n = N + 1$, σ_{uv} is the number of shortest paths from u to v, and $\sigma_{uv}(b)$ is the number passing through b. The double sum counts undirected pairs from both directions, so the denominator is $(n-1)(n-2)$ (Brandes, 2001, p. 9).

Burt's constraint. Broker ego-network constraint (Burt, 1992):

$$\text{Constraint}_b = \sum_{j \in \mathcal{N}_G(b)} \left(p_{bj} + \sum_{h \in \mathcal{N}_G(b) \setminus \{j\}} p_{bh}\, p_{hj} \right)^2$$

where $\mathcal{N}_G(u)$ is node u's set of neighbors in G, $d_u = |\mathcal{N}_G(u)|$, and $p_{uv} = 1/d_u$ when $v \in \mathcal{N}_G(u)$, and zero otherwise (Everett & Borgatti, 2020). The indirect term uses intermediary h's degree in G.

Effective size. Non-redundant broker contacts using the Borgatti (1997) binary-undirected simplification:

$$\mathrm{ES}_b = d_b - \frac{2t_b}{d_b},$$

where d_b is the broker's degree and t_b is the number of ties among the broker's neighbors, excluding ties to the broker.

Prediction Quality

Prediction quality is evaluated on random candidate pairs:

Sample	Construction	Reported measures
Random candidate pairs	Up to 100 principals with nonempty histories; for each sampled principal i, $\min(40, N-1)$ non-self partners j. Each predicted match value is evaluated against $Q + f(\mathbf{x}_i, \mathbf{x}_j)$ for the same sampled pair. Both models use the same sample, which is deterministic conditional on seed and period.	Per-principal R^2, root mean squared error (RMSE), bias, and Spearman rank correlation, averaged across sampled principals.

The definitions are

$$R^2 = 1 - \frac{\mathrm{MSE}}{\mathrm{Var}_{\mathrm{pop}}(q)}, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum(\hat{q} - q)^2}, \qquad \mathrm{Bias} = \frac{1}{n}\sum(\hat{q} - q).$$

Rank correlation evaluates the ranking produced from predicted match value. Candidates are ordered by predicted match value, with equal predictions ordered uniformly at random. All tie-breaking draws are independent. For random candidate pairs, Spearman's ρ_S compares this produced ranking with ranks of underlying match value. Thus a constant predictor would produce a random ranking with expected rank correlation zero. In the definitions above, q denotes the relevant target: underlying match value for random candidate pairs. Random-candidate summaries exclude a principal's candidate set from all four measures if it contains fewer than five pairs or if the population variance of its underlying match values is below $\sigma_\varepsilon^2/6$. The R^2, RMSE, bias, and rank-correlation means therefore use the same valid-principal sample.

Other Measures

Measure	Definition
Access vs. assessment	For each accepted broker-directed offer, record whether receiver j was a direct neighbor of sender i in G at the time of the offer. Non-neighbor receivers indicate access; neighbor receivers indicate assessment.
Realized match output by channel	Average realized match output $\bar{q}_c^t$, where $c \in \{\text{self}, \text{brokered}\}$.
Requested-position fill rates	For each channel, accepted directed offers divided by requested positions.
Net output per principal	Total realized output from completed relationships, counting each relationship once, minus all self-search costs and broker fees, divided by N. Self-search costs apply to every self-search request, successful or not. Broker fees apply to each accepted broker-directed offer, including both offers when a reciprocal relationship fulfills two brokered requests.
Mean channel satisfaction	Cross-principal means $N^{-1}\sum_i s_{i,\text{self}}^t$ and $N^{-1}\sum_i s_{i,\text{broker}}^t$.
Outsourcing rate	Outsourced requested positions divided by total requested positions.
Broker access size	$\lvert\mathcal{A}_b^t\rvert = \lvert\text{Roster}^t \cup D^t\rvert$, measured after outsourcing decisions form D^t and before entry/exit.
Counterparty concentration	Median and maximum current-period counterparties per principal.
Principal-node degree summaries	Mean, median, minimum, and maximum degree across principals.

Part II. Baseline Values and Reproducibility

9. Baseline Values and Reproducibility

9a. Baseline Parameter Values

The table lists baseline values only. Entries are arranged in two columns; each column follows the document order, with the run length listed first.

The baseline learning function is NN. Ridge and its broker variants are alternate experiments (§3).

The baseline broker provides both access and assessment. The assessment-only and access-only modes are alternate experiments (§6c).

Symbol	Baseline	Defined in	Symbol	Baseline	Defined in
T	500	§1	f_r	0.60	§4a
N	1000	§1	λ_c	0.05	§4c
d	8	§1	k_G	6	§5a
d_γ	8	§1	p_{rewire}	0.1	§5a
σ_x	0.5	§1	η	0.02	§5c
Q	2.84	§2a	K	5	§6
ρ	0.50	§2a	p_{demand}	0.50	§6
δ	0.5	§2c	$n_{\text{strangers}}$	10	§6a
σ_ε	0.28	§2a	α_R	0.20	§6b
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	0.003, 0.03	§3a	p_{roster}	0.02	§6b
$E_{\text{init},a}, E_{\text{init},b}$	100, 100	§3a	ω	0.2	§7a
$E_{\text{steps},a}, E_{\text{steps},b}$	50, 50	§3a	Δ_{net}	20	§8
W_p	40 periods	§3a	M_{train}	2000	§3a

9b. Reproducibility

All model-event randomness flows from a single integer seed. The seed determines type draws, G, matching function parameters ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{B}$), broker seed roster, standing-roster churn and replenishment, random ordering within market score ties, and subsequent model events. Random candidate samples and their independent principal and broker tie-breaking draws are deterministic conditional on seed and period. Simulations are fully reproducible given parameters and seed.

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Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of 'small-world' networks. *Nature*, *393*(6684), 440–442.

Supplementary Material

Simulation design and analysis

Main sweeps. The main simulations use neural-network learning and run for $T = 500$ periods. Table S1 gives the parameters varied one at a time; Table S2 gives the crossed designs. Parameters outside each sweep remain at baseline (Appendix B, §9). The reservation coefficient f_r sets the threshold $r = f_r \bar{q}_{\text{cal}}$.

Table S1: Baseline values and one-at-a-time sweeps.

Parameter	Baseline	Values varied
Matching composition, ρ	0.5	0, 0.15, 0.3, 0.5, 0.7, 0.85, 1
Turnover probability, η	0.02	0, 0.001, 0.01, 0.02, 0.03
Number of principals, N	1000	500, 1000, 1500
Reservation coefficient, f_r	0.6	0.4, 0.6, 0.9, 1.2
Matching difficulty, δ	0.5	0, 0.25, 0.5, 0.75, 1
Initial ring degree, k_G	6	4, 6, 12
Broker roster share, α_R	0.2	0.1, 0.2, 0.4
Self-search stranger pool, $n_{\text{strangers}}$	10	0, 10, 50

Table S2: Crossed designs. Each row includes all combinations of the listed values.

Parameters	First parameter	Second parameter
$\rho \times \eta$	0, 0.15, 0.5, 0.85, 1	0, 0.001, 0.01, 0.02, 0.03
$\rho \times f_r$	0, 0.5, 1	0.4, 0.6, 0.9, 1.2
$\eta \times f_r$	0, 0.001, 0.01, 0.02, 0.03	0.4, 0.6, 0.9, 1.2
$\rho \times \delta$	0, 0.15, 0.3, 0.5, 0.7, 0.85, 1	0, 0.25, 0.5, 0.75, 1

The sweeps contain 80 distinct regimes. Each uses seeds 1–20; the baseline uses seeds 1–50, giving 1,630 runs per learning model. Configurations shared by sweeps use the same runs. At $\rho = 1$, δ has no effect, so its grid coordinates share one regime.

Learning comparisons. The complete design is repeated with ridge regression. Restrictions on the ridge broker assess the contributions of observation volume and information about both parties: a fitting sample capped at the median principal sample size, one party recorded per match, and omission of the interaction term. Each restriction is compared with the unrestricted ridge broker across the 31 distinct regimes in the $\rho \times \delta$ grid, including the baseline. Each restriction uses 650 runs with the same seed allocation as the main sweeps. Principal learning rules are unchanged.

Broker services. The neural-network model compares full service, assessment-only service, and access-only service. The comparison crosses $\rho \in \{0, 0.5, 1\}$ with $\eta \in \{0, 0.01, 0.03\}$, and adds $(\rho, \eta) = (0.5, 0.001)$ and $(\rho, \eta) = (0.5, 0.02)$. Other parameters remain at baseline. Each service uses 50 seeds at the baseline and 20 elsewhere: 11 market regimes and 750 runs across all services.

Training calibration. Separate baseline calibration runs select neural-network training settings. The screen varies learning rates $\{0.003, 0.01, 0.03\}$ and recurrent update budgets $\{50, 100, 200\}$, one

learner at a time, with initial budgets twice the recurrent budget. It uses 17 configurations, 3 seeds per configuration, and 200 periods. Selection considers each learner's ranking performance, its stability across periods and seeds, and runtime. The selected learning rates are 0.003 for principals and 0.03 for the broker, with 100 initial and 50 recurrent steps for both. Confirmation uses 5 seeds, including the screening seeds.

Ridge calibration crosses the penalties $\{0.0003, 0.001, 0.003, 0.01, 0.03\}$ for principals and the broker, using 10 seeds per combination and 500 periods. Selection considers each learner's ranking performance, stability, and prediction error. The reported experiments use penalties of 0.001 for both learners. Confirmation uses 5 new seeds. Calibration seeds are excluded from the reporting runs.

Analysis. Early and late summaries cover periods 51–70 and 481–500. Betweenness, constraint, and effective size use snapshots at periods 60 and 500, respectively; other measures are averaged within each window. Ensemble means average these run-level summaries across seeds. Comparisons between model versions use differences paired by seed; intervals are pointwise 95% Student-t Monte Carlo confidence intervals across independent seeds. Summaries across regimes give each distinct regime equal weight. Outcome definitions and the Monte Carlo precision assessment are reported in Section IV.A.

Supplementary figures

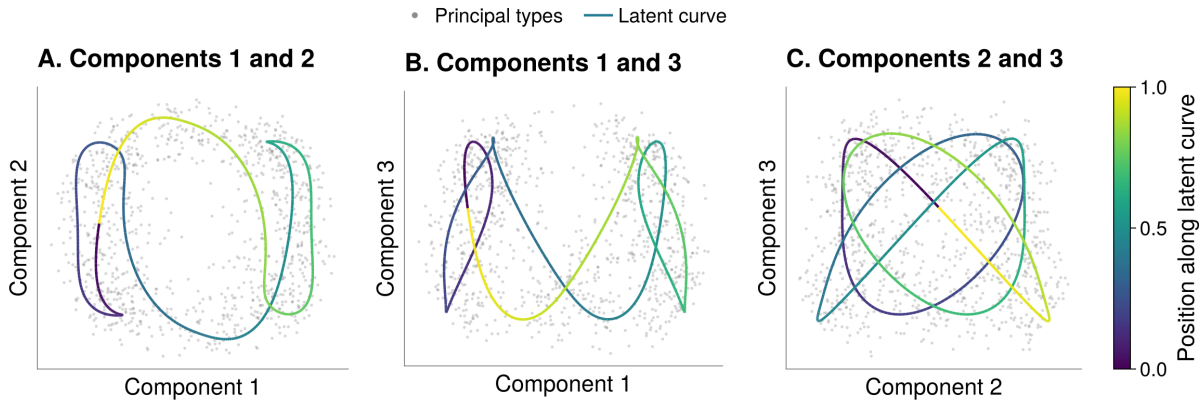


Figure S1: Principal types and their latent curve. All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.

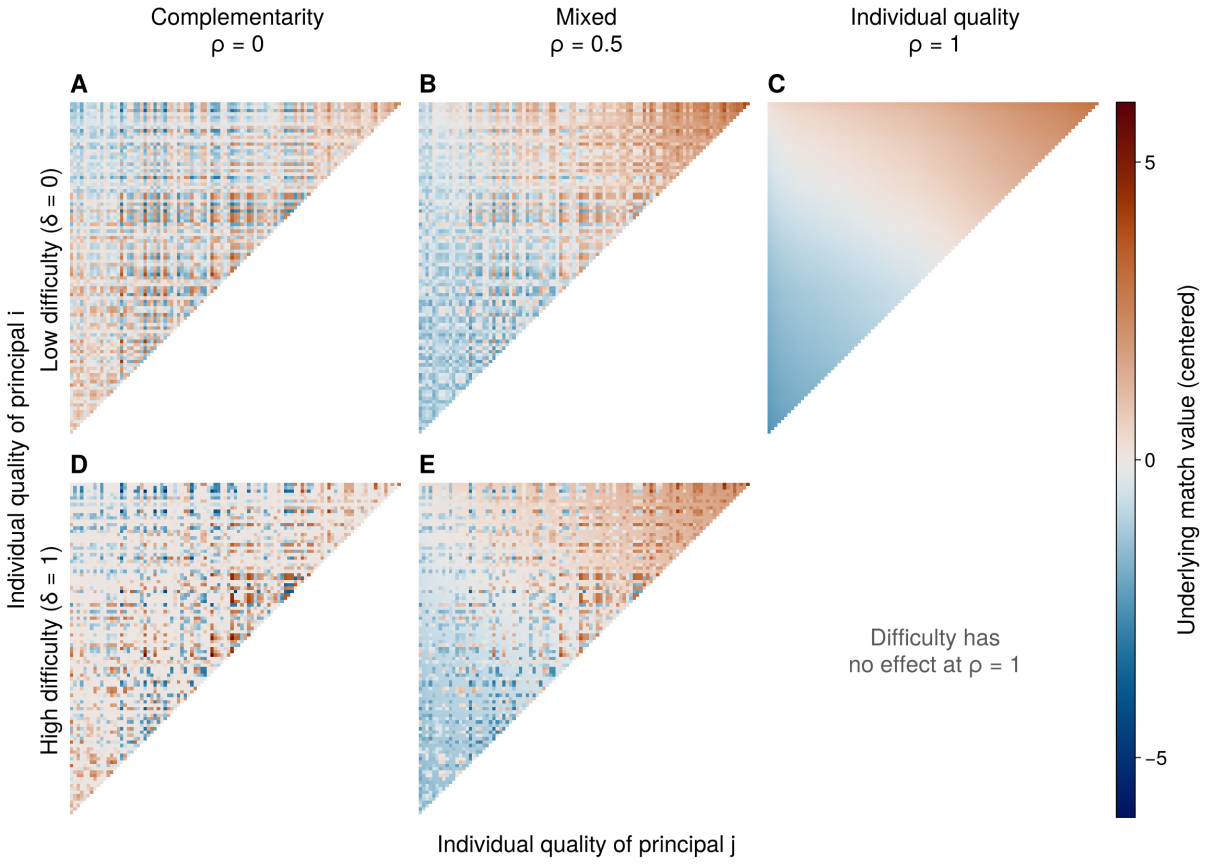


Figure S2: Underlying match value across pairs of principal types. 100 types selected at evenly spaced ranks of individual quality from realization 1 (1000 principals). Types and matching-function objects c, A, and B are fixed across panels. Individual quality increases rightward and upward. Upper triangles exclude self-matches; colors show each pair's underlying value minus the grand mean of the full value matrix.

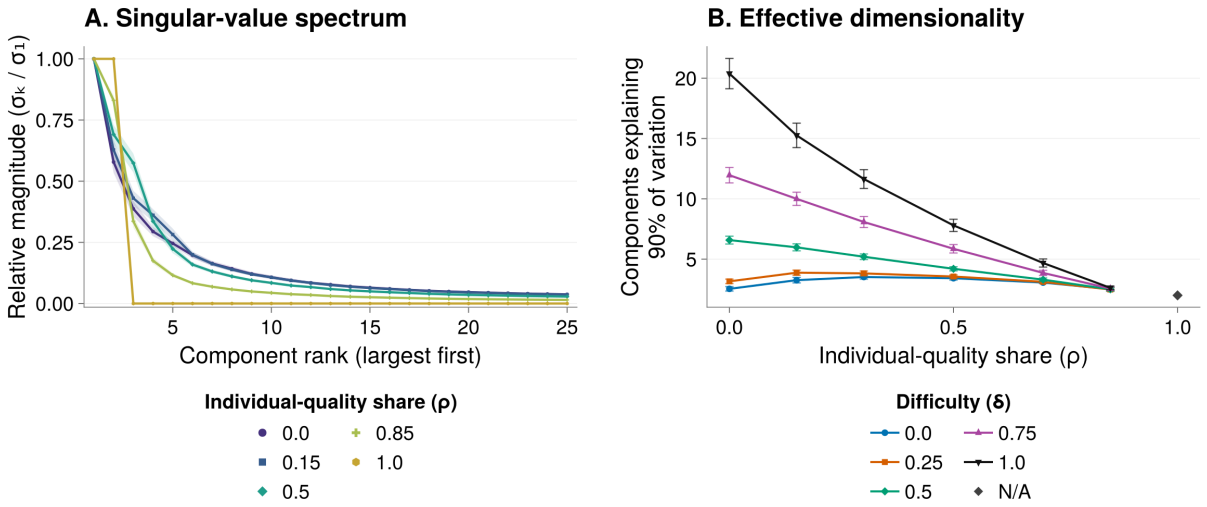


Figure S3: Dimensionality of underlying match value. Matrices of pair-specific underlying match values, centered within each realization and summarized over 50 independent realizations. Types and matching-function draws are fixed within each realization as ρ and δ vary. A: the first 25 singular values at $\delta = 0.5$, divided by the largest within each realization. B: the fewest components accounting for 90% of the sum of squared singular values. Lines and markers are means; ribbons and whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where underlying match value depends only on individual quality and difficulty is inapplicable.

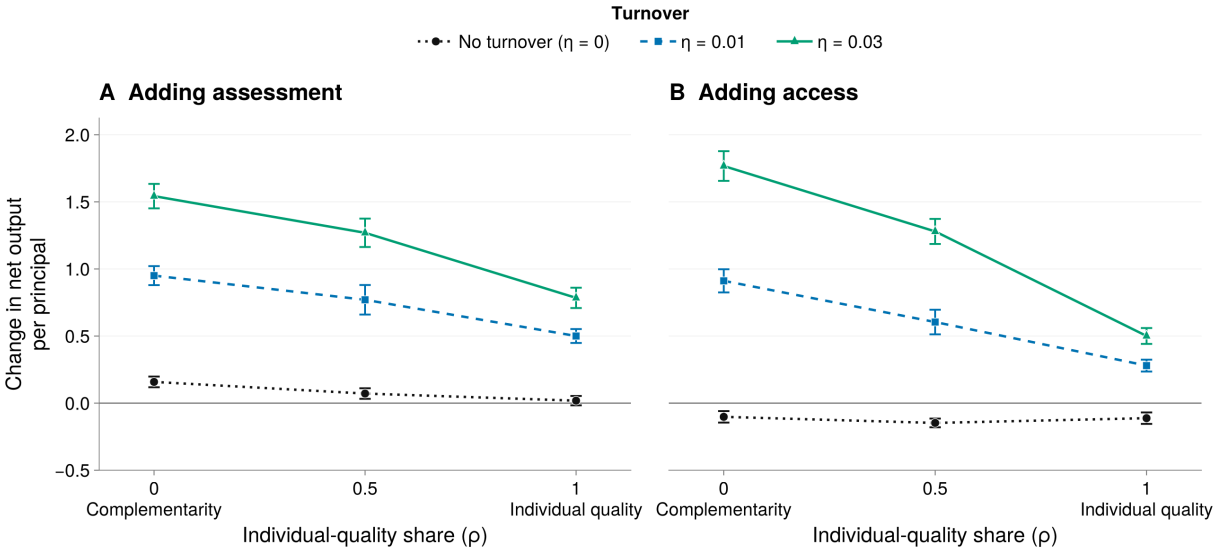


Figure S4: Contributions of assessment and access across matching composition. Late-mean differences paired by seed: A, full service minus access-only; B, full service minus assessment-only. Parameters other than matching composition and turnover remain at baseline. Points average 20 seeds per regime; bars show pointwise 95% Monte Carlo intervals. Connecting lines are visual guides.

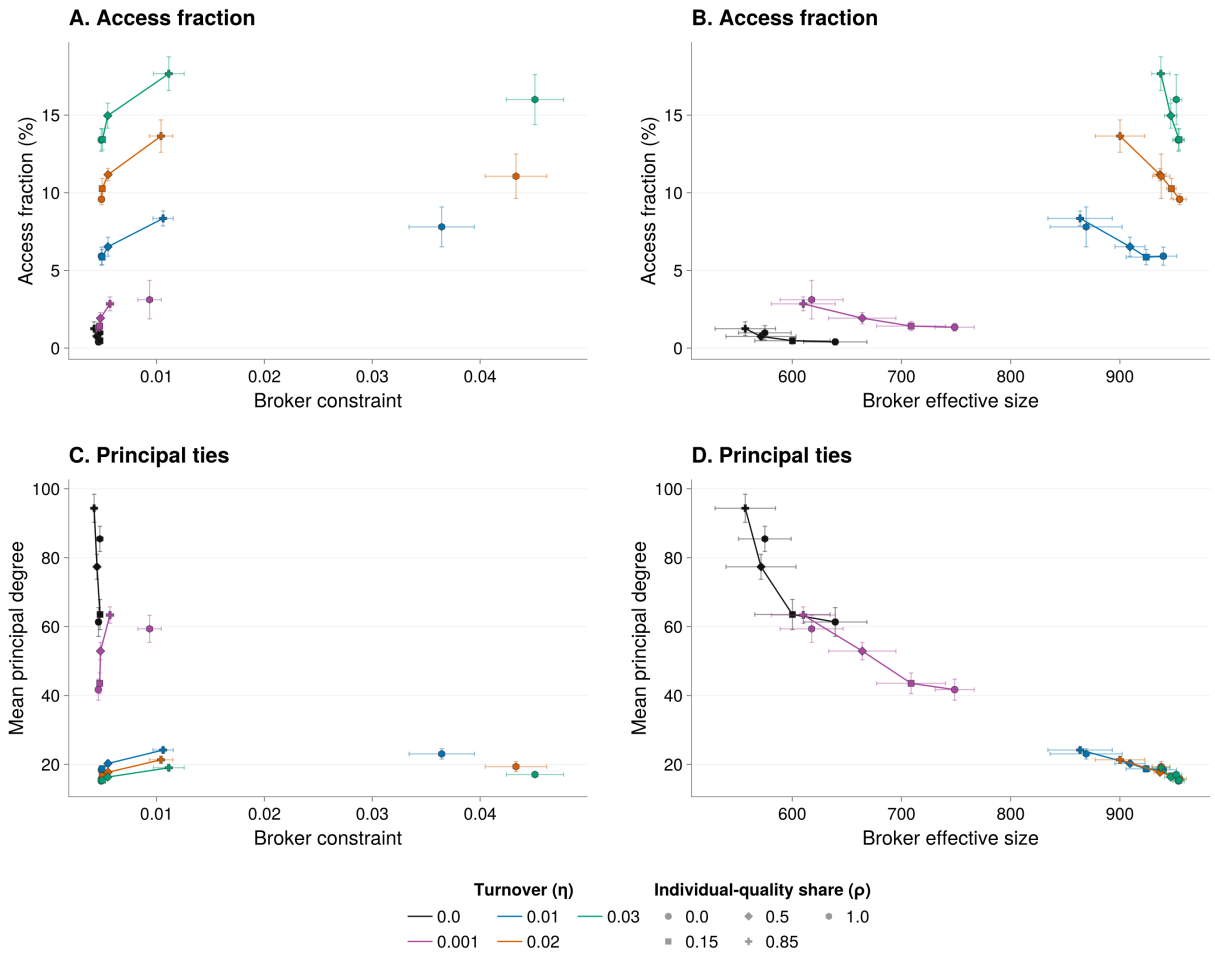


Figure S5: Structural position and access under alternative measures. Ensemble means from the neural-network simulations, using constraint and effective size at period 500 and late-window averages for access and degree. Parameters other than matching composition and turnover remain at baseline. Lines connect matching compositions at fixed turnover, with $\rho = 1$ shown separately. Principal degree includes the broker tie when present. Whiskers show pointwise 95% Monte Carlo intervals.

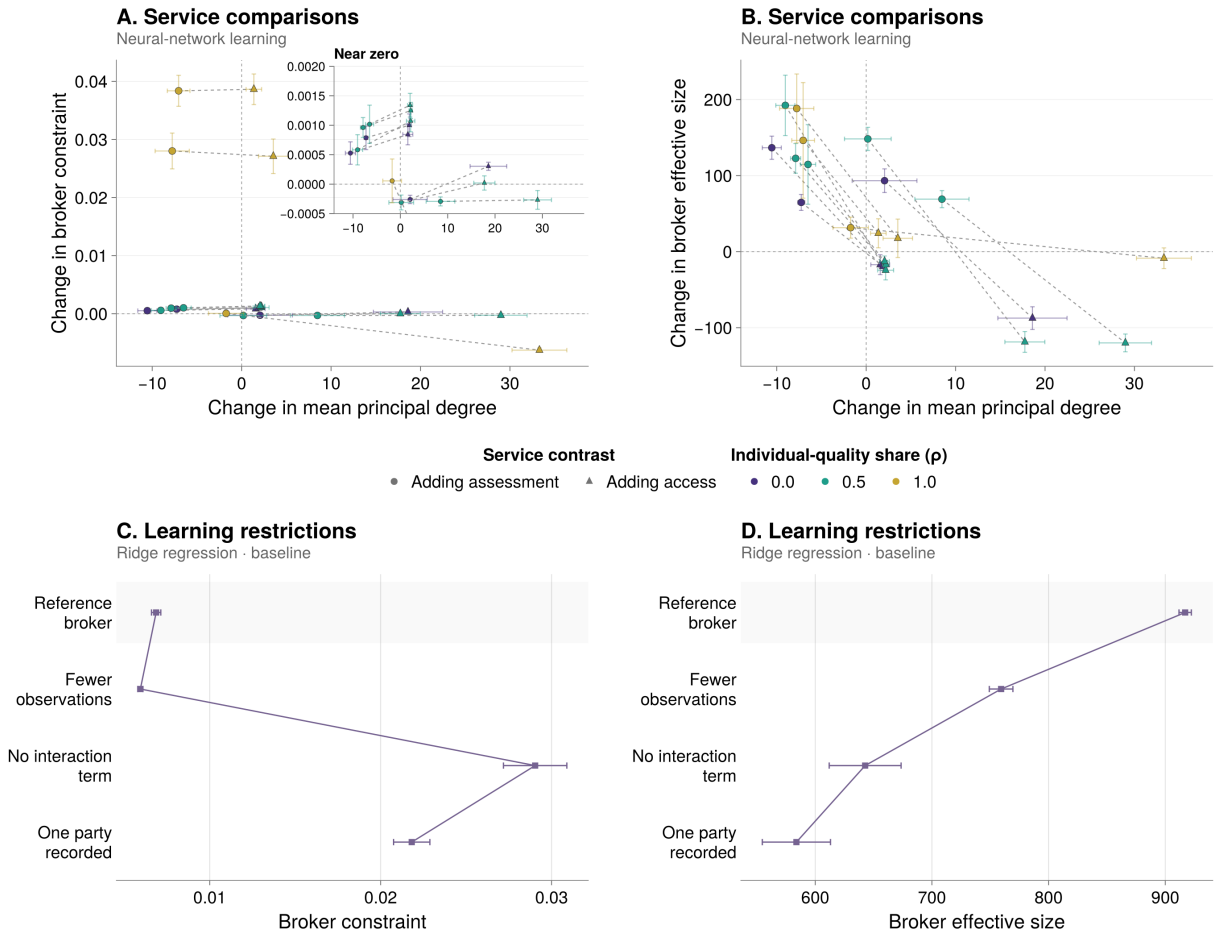


Figure S6: Structural consequences of services and learning under alternative measures. A–B: differences paired by seed across matching composition and turnover, with other parameters at baseline. Adding assessment is full service minus access-only; adding access is full service minus assessment-only. Dashed lines connect contrasts within each regime; the inset enlarges the region near zero. C–D: ensemble means from the baseline ridge experiments. Constraint and effective size use period 500; other measures use late-window averages. Whiskers show pointwise 95% Monte Carlo intervals.

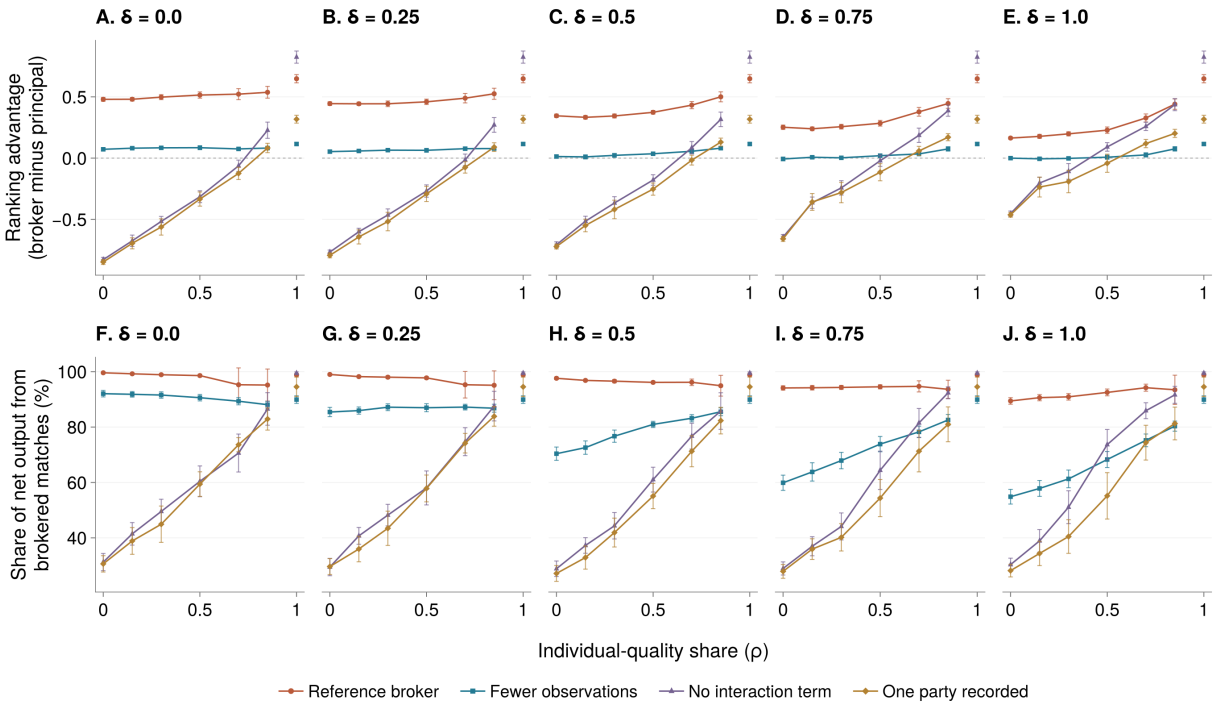


Figure S7: Sources of the broker's advantage across matching difficulties. Ensemble late means from the ridge experiments; parameters other than ρ and δ remain at baseline. Ranking differences and net-output shares are calculated within seed before averaging. Whiskers show pointwise 95% Monte Carlo intervals. The same $\rho = 1$ estimates appear in each column because difficulty has no effect there. Lines are visual guides.